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CMP6214 User Experience Design

SignSpeak (A Sign Language Translation App)

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1 Introduction

1.1 Concept

The concept of this project is a **real-time sign language translation mobile application** that will assist a wearable **surface electromyography (sEMG) and inertial measurement unit (IMU) wristband** to translate sign language gestures into text and speech thus reducing/eliminating a need for external assistance. The application is designed to support smoother communication between sign language users and individuals who do not understand sign language particularly in everyday, informal and spontaneous interactions. From a user experience perspective, the app prioritises clarity, accessibility and ease of use, ensuring that users can quickly connect the wearable device, calibrate sensors, initiate translation and interpret output with minimal cognitive effort. The goal of the application is not only to provide translation functionality but also to build user trust and confidence through clear feedback, intuitive navigation and supportive error handling.

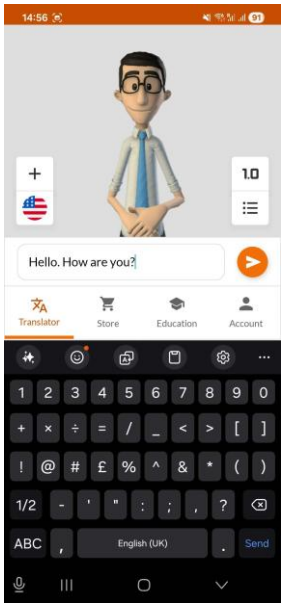
1.2 Target Audience

The target audience for this application is primarily the deaf and hard-of-hearing sign language users, but it is also made accessible and usable for non-sign language users like family members, friends, educators etc. These users may vary in age, technical experience, and communication preferences, making accessibility and simplicity critical design considerations. As sign language is a visual and gesture-based form of communication, the interface is designed to prioritise visual clarity, readable text and clear system feedback. By considering both user sets, the application aims to function as a bridge between two user groups, supporting inclusive communication without requiring either party to possess specialist knowledge or technical expertise.

1.3 Market Research and Competitor Analysis

While exploring similar mobile applications, I concentrated on pinpointing solutions that were not only practical but also user-friendly and accessible from the standpoint of user experience. Instead of discovering a single application that perfectly aligned with my concept, I scrutinized how current applications handle gesture input, translation output, onboarding processes and accessibility features. The majority of applications related to sign language can be classified into two primary categories: those focused on translation or learning tools and those aimed at providing speech or text-based accessibility. This project aspires to integrate effective user experience elements from these existing solutions while simultaneously addressing their shortcomings in real-time communication.

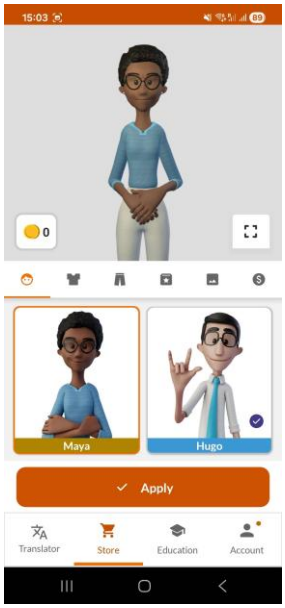
A pertinent mobile application is **Hand Talk: ASL Sign Language**, which is available on both iOS (https://apps.apple.com/us/app/hand-talk-asl-sign-language/id659816995) and Android (https://play.google.com/store/apps/details?id=br.com.handtalk&hl=en). This application translates text and speech into sign language through the use of animated characters/models and incorporates educational features such as quizzes and guided lessons. Its interface is visually straightforward and easy to navigate, rendering it effective for both learning and reference purposes.



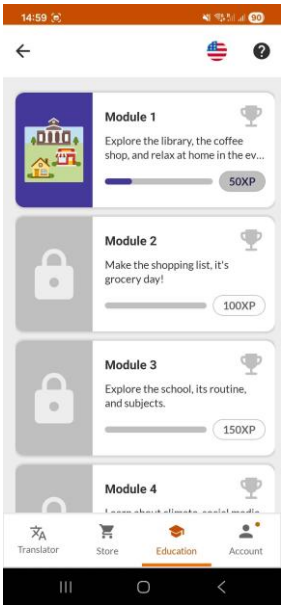
Input for Model to Sign



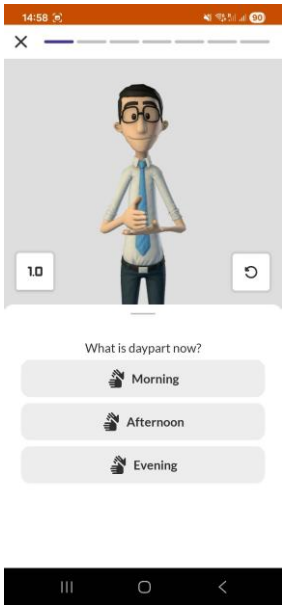
Model Signing the Input



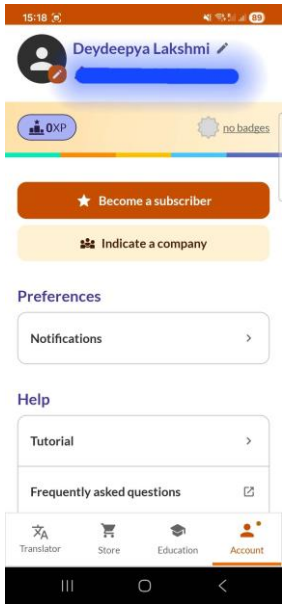
Choose Model of Liking



Learning Section



Quiz in One of the Modules



Account Settings Section

Figure 1: Hand Talk App

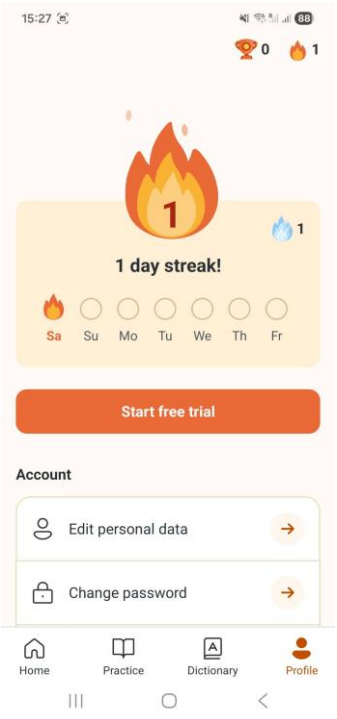
However, while Hand Talk performs well as an educational and translation-from-text tool, it lacks real-time sign language detection. This means users can learn signs or view translations, but the app cannot support spontaneous, day-to-day communication through gesture input.

Another application taken into consideration is **Lingvano**, a mobile app available on both iOS (https://apps.apple.com/vn/app/lingvano-learn-sign-language/id1547252782?l=vi) and Android (https://play.google.com/store/apps/details?id=com.lingvano.app). Lingvano is designed as a sign language learning platform, offering short, structured lessons that include videos of real signers, interactive exercises and progress tracking. From a user experience perspective, the app uses a clean layout, clear visual hierarchy and guided onboarding,

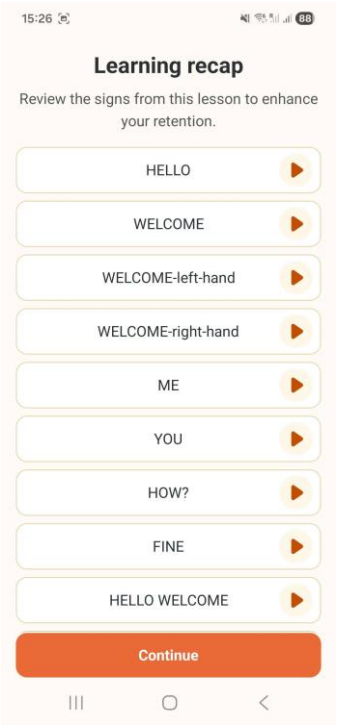
making it easy for users to understand how to navigate the app and begin learning with minimal effort.



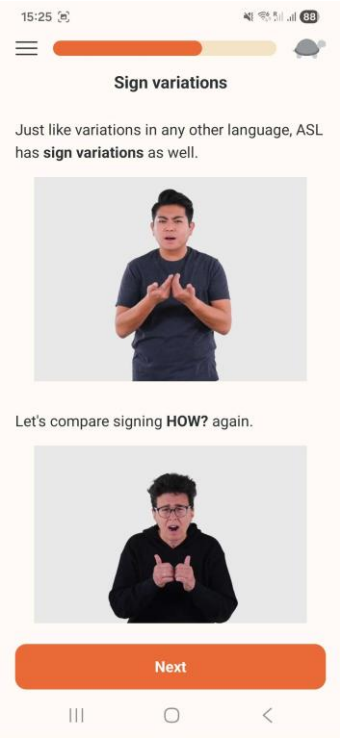
Learning Units



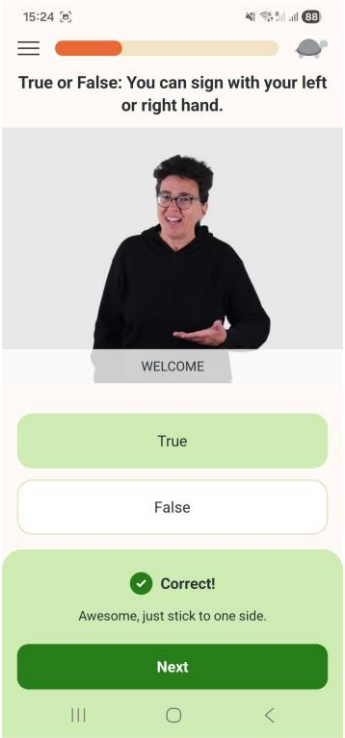
Duolingo-like Streak Documented to Motivate Learning



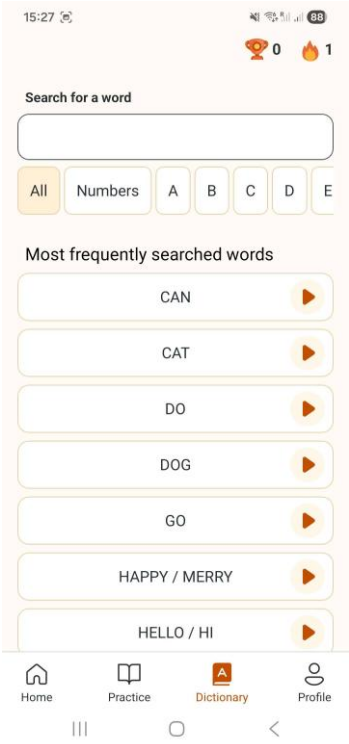
Recap of Signs Learnt



Demo of a Class Within Module



Quiz



Dictionary

Figure 2: Lingvano

However, Lingvano is focused on learning rather than real-time communication and while it performs well in terms of learnability, consistency and feedback during practice quizzes, it doesn't have camera detection capabilities to actually check if you're signing accurately. This along with the fact that it is not built to translate conversations, limits its usefulness in spontaneous, day-to-day communication scenarios.
The app's strength however is in structured onboarding, clear instructions and immediate feedback highlighting important UX principles that inform this project, while its limitations further emphasise the need for a solution that supports natural sign language interaction alongside an intuitive and accessible interface.

In summary,

App	UX Strengths (+)	UX Limitations (–)	UX Takeaways for This Project
Hand Talk	Clear interface, simple navigation, engaging visual output	No real-time sign detection and thus limited for live communication	Importance of clear output and accessible UI
Lingvano	Strong onboarding, structured lessons, clear feedback based on quiz performances	Learning-focused, no live translation and doesn't have a model checking if you're signing correctly	Value of guided flows and immediate feedback in concepts

Table 1: Conclusions from Market Research

2 User Experience Design

This project follows a user-centered design (UCD) approach where users and their needs are at the very core of every design decision that as defined by the Interaction Design Foundation (n.d.), is an iterative process that focuses on understanding users, defining their requirements, designing solutions and evaluating those solutions through continuous refinement.

2.1 User Personas and Respective Scenarios

User personas and scenarios are used to represent key target user groups and to demonstrate how the proposed system would be used in realistic, everyday contexts. Personas help designers maintain a user-centred perspective, while scenarios illustrate goals, motivations, and pain points in action, supporting informed design decisions (Interaction Design Foundation, n.d.).

Persona 1: Liz Augustenberg

Liz Augustenberg is a Deaf ASL user and a senior graphic designer based in Magdeburg, Germany. Her primary goal is to communicate naturally with hearing colleagues and the public without relying on interpreters or written back-and-forth communication. She values spontaneity in social and professional interactions and strives to be as independent as any of us can be in day-to-day interactions.

Liz experiences frustration with existing camera-based sign language recognition systems, which require stable lighting, fixed positioning and continuous line-of-sight. These constraints make them impractical in real-world environments. Writing messages on a phone is slow and interrupts conversational flow, while video relay services feel formal and dependent on strong internet connectivity. As a technologically confident user who regularly uses accessibility tools and wearable devices, Liz is comfortable adopting new technology if it provides clear feedback, reliability and minimal cognitive load. Her scenario highlights the need for discreet, real-time translation with immediate visual and haptic feedback in everyday contexts.

Scenario - Everyday Social Interaction (Morning Coffee Order):
While ordering coffee at a busy café, Liz uses the wearable system paired with her mobile app to sign naturally. Her muscle movements are captured and translated into spoken output in real time, allowing her to place her order without typing or showing her phone. The app provides subtle visual and haptic feedback to confirm successful translation. This scenario highlights the need for real-time translation, minimal cognitive load and discreet feedback in everyday public interactions.

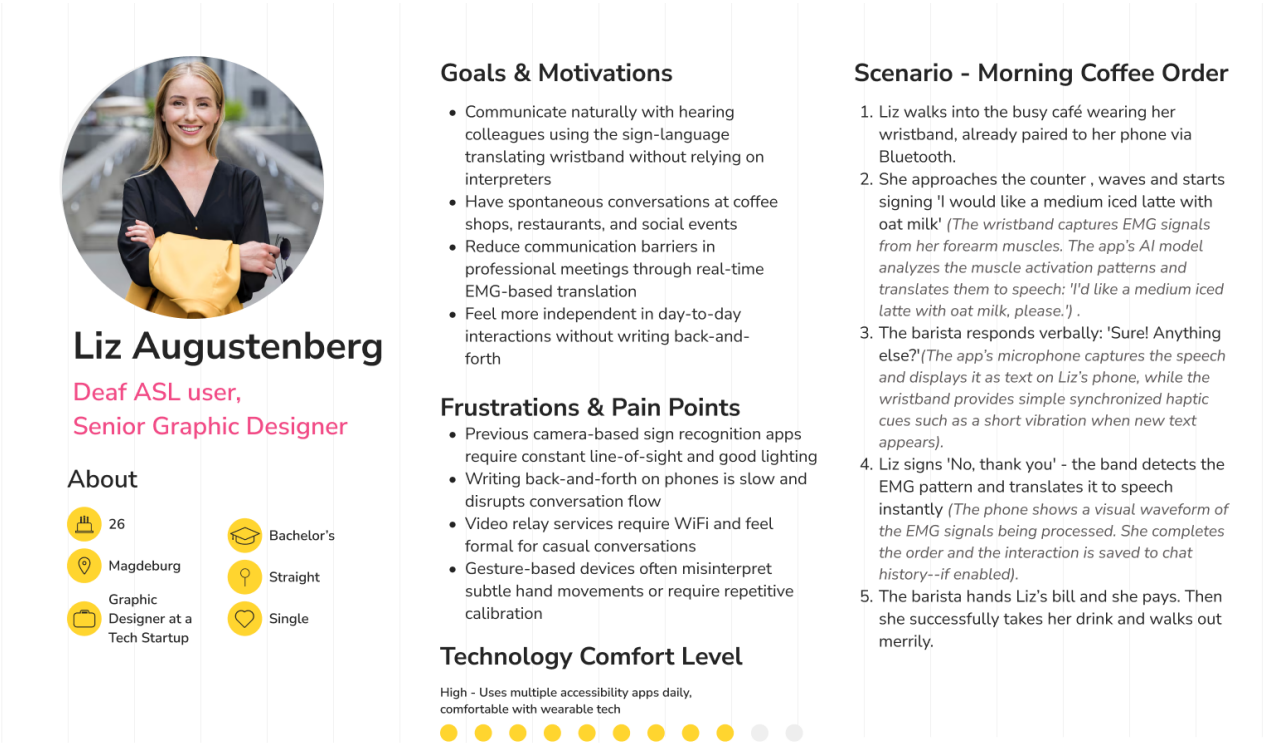


Figure 3: User Persona 1

Persona 2: Marcus Singh

Marcus Singh is a London based HR manager working in a healthcare firm with a deaf daughter. His main motivation is to learn ASL effectively in order to communicate better with his daughter and engage more confidently with the Deaf community. He is particularly interested in understanding how muscle movements correspond to signs so he can improve his accuracy and confidence.

Marcus finds existing learning methods limiting, as video-based tutorials do not provide personalised feedback on signing accuracy. Camera-based applications require precise positioning and do not clearly indicate whether gestures are being performed correctly. His busy work schedule also makes it difficult to attend structured ASL classes. With a medium level of technological comfort, Marcus benefits from guided learning, clear instructions and immediate feedback. His scenario emphasises the importance of supportive onboarding, practice modes and progress visualisation to build confidence and motivation in learning sign language.

Scenario – Evening Learning Session:
After work, Marcus opens the app's learning mode to practise ASL. The system guides him through signing exercises, analysing muscle activation patterns and providing real-time feedback on accuracy. Visual progress indicators show improvement over time, helping Marcus build confidence. This scenario emphasises the importance of guided practice, feedback-driven learning and progress visualisation to support long-term skill development.

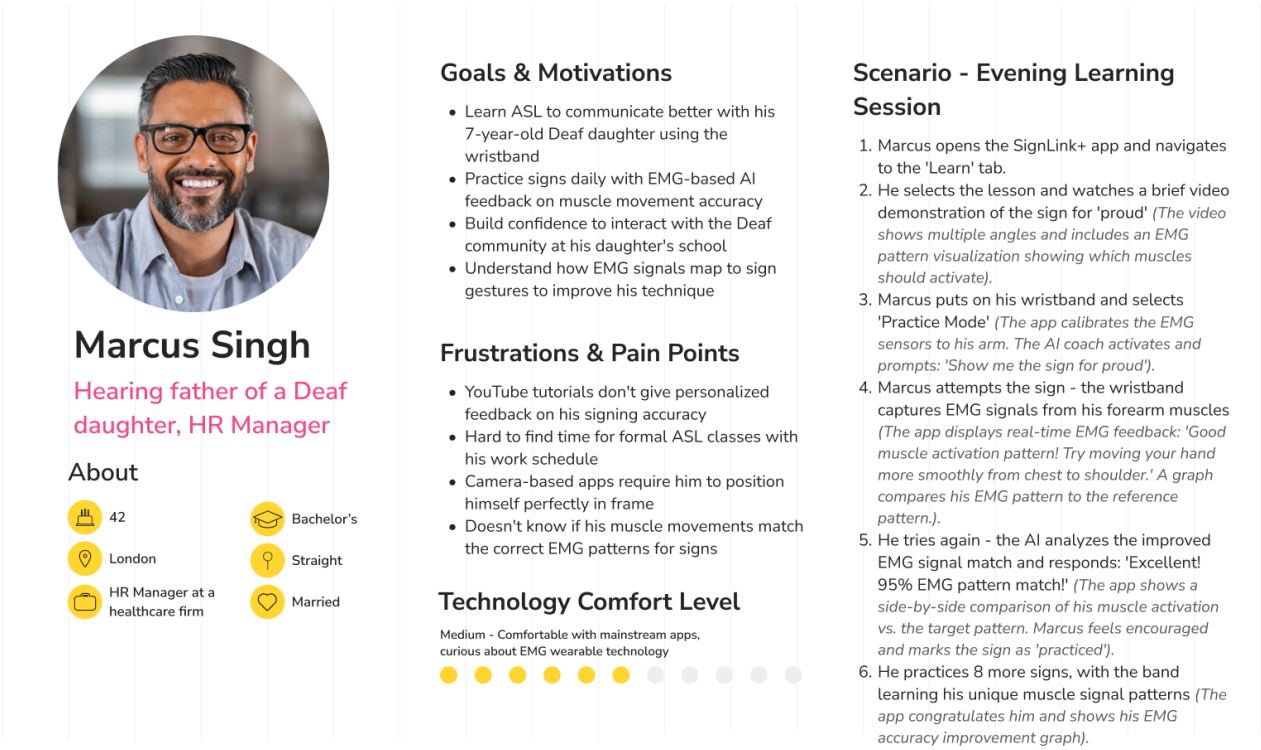


Figure 4: User Persona 2

Persona 3: Raven Nguyen

Dr. Raven Nguyen is a Hanoi based family physician working in Hallstavik, Sweden who requires accurate and reliable communication tools in clinical settings. Her primary goal is to communicate effectively with Deaf patients without delays caused by interpreter availability, while also ensuring privacy and compliance with healthcare regulations. She values discreet, hands-free interaction that does not interfere with patient consultations. Her frustrations include the limited availability of interpreters, complex medical terminology that is poorly supported in existing camera-based translation tools and privacy concerns related to video recording. As an early adopter of health technology and wearable devices,

Dr. Nguyen is highly comfortable with advanced systems, provided they are secure and reliable. Her scenario demonstrates the need for precise translation, domain-specific language support and privacy-focused design, reinforcing the importance of trust, transparency and system feedback in high-stakes environments.

Scenario – Patient Consultation:
During a consultation with a Deaf patient, Dr. Nguyen uses the wearable translation system to enable real-time communication without recording video or audio. Medical questions and responses are translated accurately and discreetly, allowing the consultation to proceed smoothly. This scenario demonstrates the need for domain-specific vocabulary support, privacy-first design and high system reliability in high-stakes healthcare contexts.

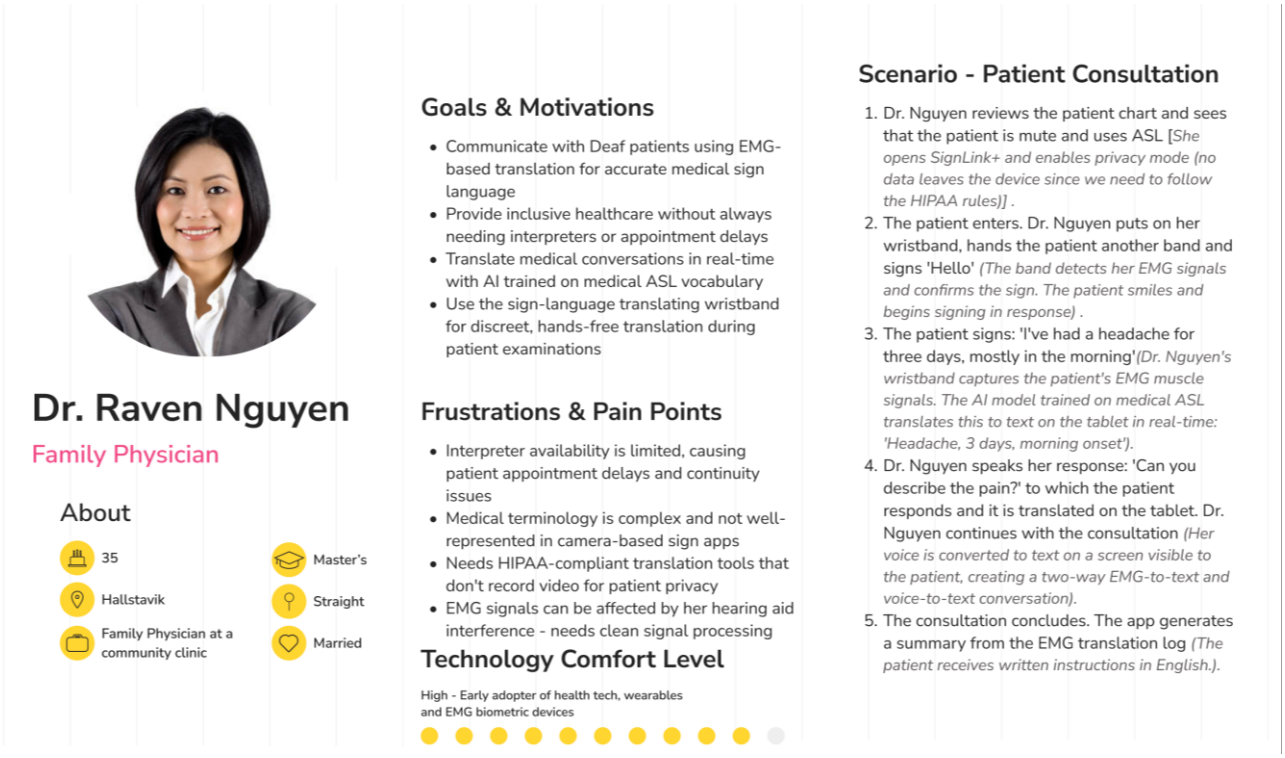


Figure 5: User Persona 3

2.2 User Requirements

Based on the personas and scenarios outlined in Section 2.1, user requirements were identified and prioritised using the **MoSCoW method**, which categorises requirements into ***M**ust Have*, ***S**hould Have*, ***C**ould Have*, and ***W**on't Have*. This approach helps ensure that essential functionality is delivered first while allowing flexibility for future enhancements. SignSpeak 's requirements are as below,

Must Have

Essential (free) features required for basic app functionality and user accessibility.

- User Authentication & Profile Management
  - Sign-up/login (email, Google, or Apple)
  - Profile includes user type (Deaf, hearing, learner, healthcare, etc.)
  - Pair and calibrate sign-language translating wristband to the app
- Real-time EMG-to-Text Translation
  - Capture and process EMG signals from band sensors
  - Display recognized sign as text on-screen
  - Simple haptic feedback on band to confirm recognition
- Text-to-Speech Output
  - Convert translated text to spoken audio in a default voice (English)
  - Basic offline speech synthesis
- Speech-to-Text Input
  - Capture microphone input and display it as text for the Deaf user
  - Trigger single vibration alert on band when new speech text appears
- Basic Chat History
  - Store last few conversations locally for reference
- Accessibility Options
  - Adjustable text size, color contrast, vibration strength

Should Have

Important but not strictly essential; adds comfort and usability.

- Basic Multi-language Support
  - Option to switch translation output language (e.g., English, Spanish, French)
  - Basic AI translation accuracy tuning
- Real-time EMG Visualization
  - Display simple waveform or graph of detected EMG signals for learning feedback (learning feature is premium)
- Tutorial Mode
  - Built-in guide for band calibration and basic signing gestures
- Offline Mode (limited)
  - EMG-to-text still works without internet; AI speech translation limited to cached vocabulary

Could Have

These are premium features that would significantly boost the project.

- Advanced AI Sign Learning Mode
  - Personalized model training on user's unique EMG signal patterns
  - Accuracy feedback for the learning sign language feature.
- Extended Multi-language & Voice Library
  - Choose voice type (male/female/neutral)
  - Translate speech into multiple languages (e.g., English → Vietnamese, etc.)
- Contextual AI Translation
  - Context-aware sentence formation for smoother spoken output
  - Emotion tone adjustment in speech synthesis
- Conversation Logging & Cloud Sync
  - Securely store, label, and access full conversation history across devices
- Customizable Haptic Feedback Patterns
  - Define vibration patterns for different types of alerts (e.g., message received, translation complete)
- Accessibility Personalization Suite
  - AI adjusts display & vibration settings based on ambient conditions (light, noise, etc.)

Won't Have

Not in the current plan (of execution) but is a future prospect.

- Gesture-only interface for full navigation (no phone use required)
- VR/AR integration for 3D ASL practice
- Myo dual-band mode (two armbands for full hand gesture tracking)
- Advanced EEG integration for cognitive/ emotional context detection

Figure 6: User Requirements

2.3 Hierarchical Task Analysis

Hierarchical Task Analysis (HTA) was used to decompose the core user goals of the application into structured tasks and subtasks. HTA helps designers understand how users achieve objectives within a system by breaking complex interactions into manageable steps, thereby informing clear information architecture and efficient task flows (Stanton et al., 2005). The HTA for this project is primarily informed by the *Must Have* user requirements defined in Section 2.3, ensuring that essential accessibility and communication functions are prioritised. Since the HTA chart is quite big, the quality of what is attached is drastically reduced. Therefore, here is an alternate link to view it in better quality: https://github.com/codingsanji/SignLink/blob/main/planning/HTA.png

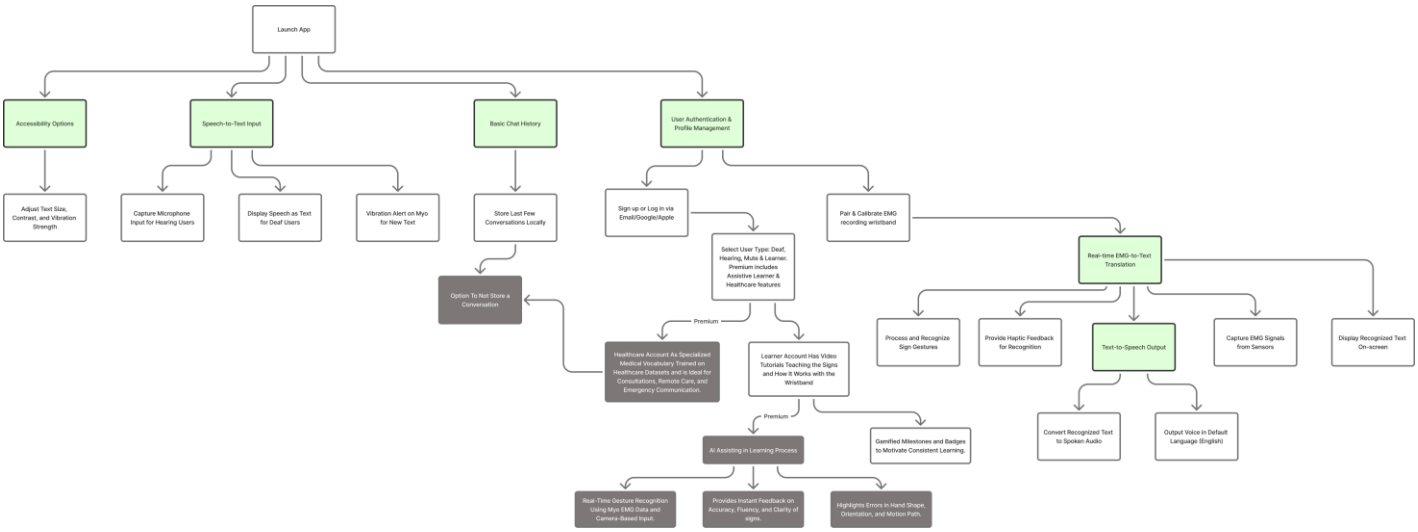


Figure 7: HTA flowchart

HTA Breakdown:

1. Launch and Set Up the Application
   - 1.1 Open application
   - 1.2 Sign up or log in (Email / Google / Apple)
   - 1.3 Select user type
     - Deaf
     - Hearing
     - Mute
     - Learner
     - (Premium: Assistive Learner / Healthcare)
   - 1.4 Pair and calibrate EMG wristband
   - 1.5 Complete initial setup

This flow ensures first-time users are guided through essential onboarding steps while minimising cognitive load. Early user-type selection allows the system to adapt interfaces and features to specific accessibility needs.

2. Configure Accessibility and Preferences
   - 2.1 Adjust text size
   - 2.2 Adjust colour contrast (dark and light mode)
   - 2.3 Adjust vibration strength
   - 2.4 Enable or disable conversation storage

Accessibility options are placed early in the hierarchy to reflect their importance for Deaf and neurodiverse users. Providing control over visual and haptic feedback supports inclusive design and aligns with accessibility best practices (W3C, 2018).

3. Perform Real-Time Communication
- 3.1 Capture EMG signals from wristband
- 3.2 Process and recognise sign gestures
- 3.3 Display recognised text on screen
- 3.4 Provide haptic feedback for recognition
- 3.5 Convert recognised text to speech (Text-to-Speech)
- 3.6 Capture microphone input (Speech-to-Text)
- 3.7 Display speech as text for Deaf users
- 3.8 Trigger vibration alert for new speech input

This task flow represents the core functionality of the system. Real-time feedback through visual text and haptic confirmation reinforces system transparency and user confidence during live communication.

4. Review Conversation History
- 4.1 View recent conversations
- 4.2 Store conversations locally
- 4.3 Option to not store conversations

Local-only storage was prioritised to address privacy concerns, particularly for healthcare and sensitive communication scenarios.

5. Premium or Specialised Features
- 5.1 Upgrade to premium
- 5.2 Access learner mode
    - Video tutorials
    - AI-assisted feedback
    - Gamified milestones and badges
- 5.3 Access healthcare mode
    - Medical vocabulary support
    - Secure consultation-oriented interface

Premium tasks are intentionally separated from core communication tasks to prevent interference with essential accessibility functions while still offering enhanced learning and professional use cases.

2.4 Low-Fidelity Prototypes (Paper)

Paper prototyping was used in the early design stage to rapidly explore interface layouts, task flows and interaction concepts before committing to digital design. Low-fidelity sketches allow designers to focus on structure, functionality and usability rather than visual aesthetics, making it easier to identify usability issues early in the design process (Gordon, 2021).

The paper prototypes focused on core *must-have* tasks identified through the MoSCoW analysis, including onboarding, user type selection, wristband pairing, real-time translation and accessibility controls. Navigation elements were kept minimal and consistent to reduce cognitive load, following Nielsen's principle of aesthetic and minimalist design (Nielsen,

2020). Primary actions such as “Start Translation” and “Speech-to-Text” were placed prominently to support recognition rather than recall, ensuring users could easily understand available actions without prior training.

Below are the low fidelity demos of the launch page (HTA 1.1) , settings page (HTA 2 - Configure Accessibility and Preferences) and the sign up and log in page (HTA 1.2)

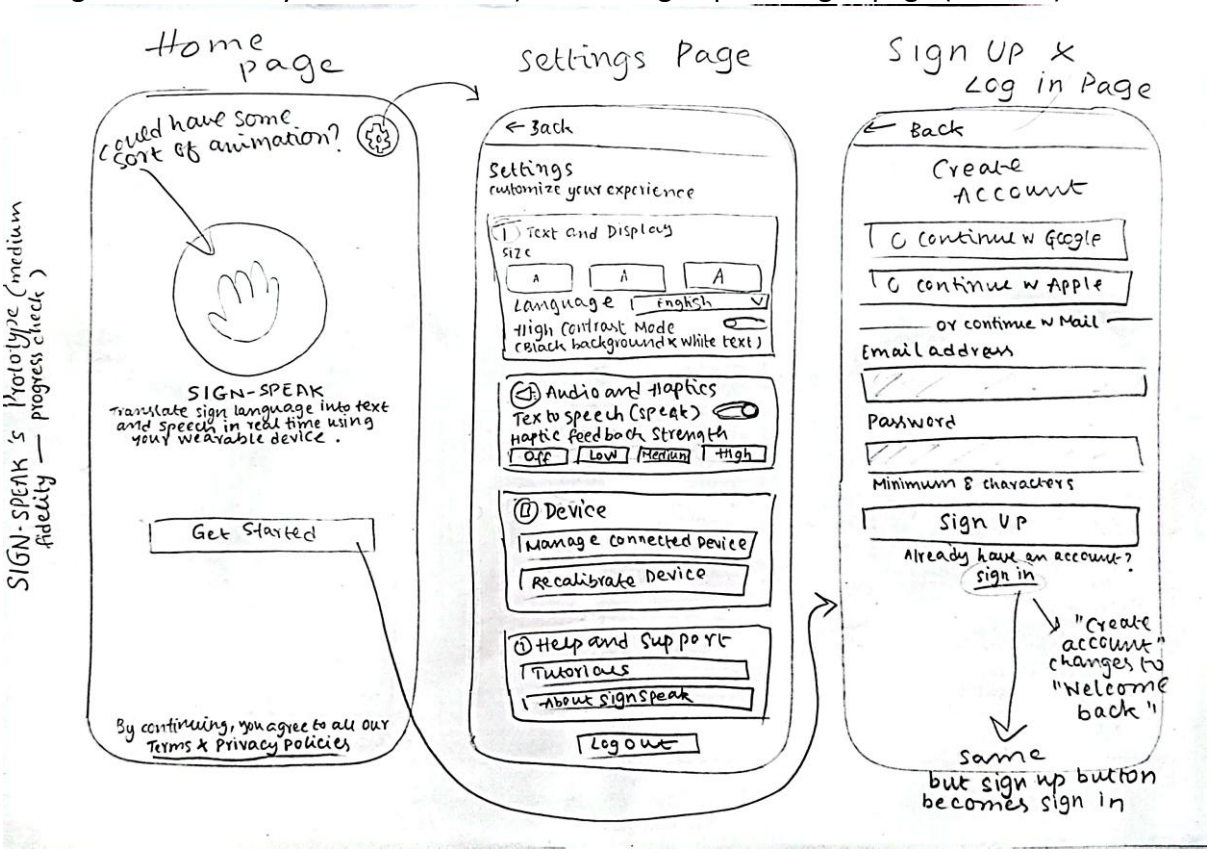


Figure 8: Section 1 of Low-fidelity prototypes

On opening the application, the first thing that will meet the user's eye is the Introduction Page. As seen in the sketch, a prominent "Settings" icon is placed at the top-right corner. This design choice allows users, particularly those with specific accessibility needs to configure the interface (e.g., high contrast or haptic feedback) before even starting the onboarding process. This supports the UX principle of User Control and Freedom, ensuring the app is usable from the very first interaction. The central call-to-action (CTA) is placed in the "thumb zone" for easy one-handed operation.

The Settings Page serves as the primary hub for HTA 2 (Configure Accessibility and Preferences). The prototype utilizes a clean, list-based layout to manage complex options without overwhelming the user.

The Sign-Up and Log-In Pages represent HTA 1.2 (Account Management), with a design focus centered on error prevention and interaction efficiency. By utilizing minimalist input fields restricted to essential data like email and password, the interface significantly reduces the cognitive "interaction cost" for the user, while clear visual headers distinguish between the "Sign Up" and "Log In" states to prevent navigation errors during the authentication phase. This account management layer is functionally vital for the SignSpeak ecosystem, as

it allows the system to securely store and retrieve the user's unique sEMG/IMU calibration profiles; by linking these biometric profiles to a specific account, the application ensures that the machine learning model recognizes the individual's specific muscle signal patterns without requiring a repetitive and tedious calibration process every time the app is opened.

Below are the low-fidelity prototypes illustrating the core operational flow, including User Type Selection (HTA 1.3), Wristband Pairing (HTA 1.4) and the Real-Time Communication Interface (HTA 3).

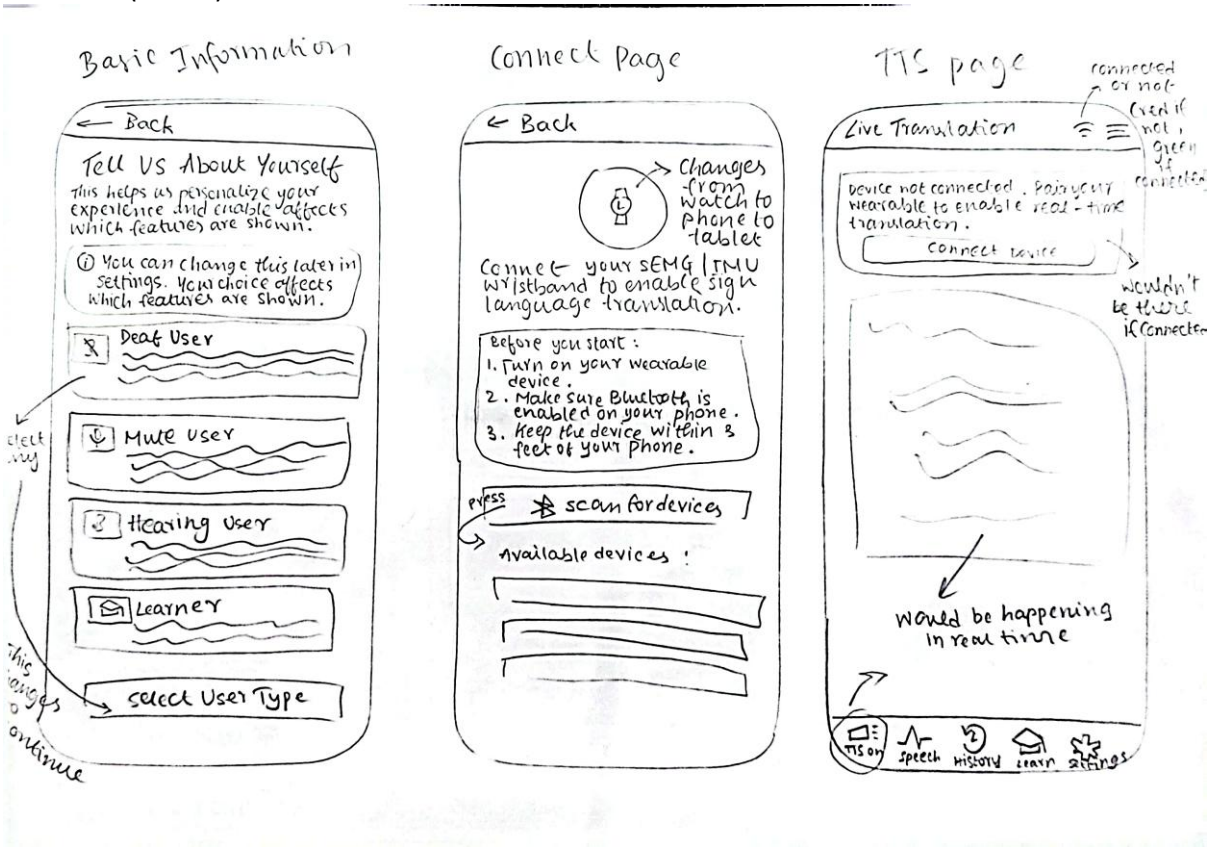


Figure 9: Section 2 of Low-fidelity prototypes

The User Type Selection screen (HTA 1.3) serves as a crucial onboarding phase that customizes the interface according to the user's designated role-"Deaf," "Hearing," or "Learner." By establishing the user's role at the outset, the application reduces cognitive burden and guarantees that accessibility features, such as haptic feedback for Deaf users or microphone input for Hearing users, are given precedence.

The Pairing screen corresponds to HTA 1.4 (Pair and calibrate EMG wristband), emphasizing the Visibility of System Status (Nielsen, 2020). The design employs a minimalist list of available Bluetooth devices alongside a conspicuous "Connect" button located within the ergonomic "thumb zone." This layout guarantees a straightforward process for establishing a hardware connection, offering clear feedback regarding the connection status prior to the user entering the more intricate translation environment.

The Main Translation Interface encompasses the fundamental functionality of the system as detailed in HTA 3 (Perform Real-Time Communication). Embracing an Aesthetic and Minimalist Design, the configuration includes a spacious text field for real-time gesture recognition (HTA 3.3) and specific toggles for Text-to-Speech (HTA 3.5) and Speech-to-Text

(HTA 3.6). This multimodal interface facilitates two-way communication, utilizing visual text and vibration alerts (HTA 3.8) to ensure a fluid exchange between signers and non-signers. The design prioritizes recognition over recall, ensuring that essential tools are readily available during live interactions.

Below is the low-fidelity prototyping section focusing on Conversation History (HTA 4) and Speech-to-Text (HTA 3.6).

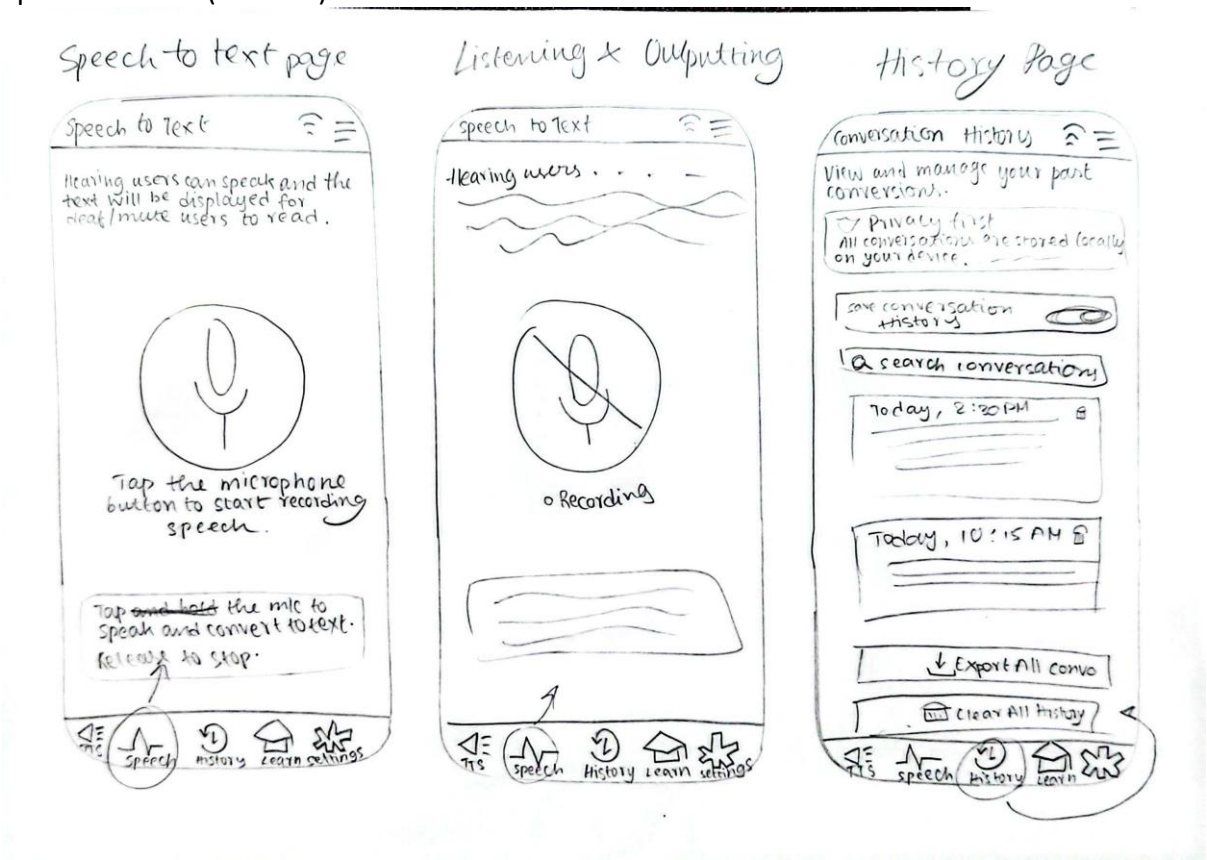


Figure 10: Section 3 of Low-fidelity prototypes

The Conversation History screen is aligned with HTA 4 (Review Conversation History). In order to mitigate the privacy issues commonly associated with healthcare and sensitive communication contexts, the interface has been crafted to prioritize local-only storage (HTA 4.2). The design features a straightforward, chronological list of previous interactions, enabling users to swiftly access their recent conversations (HTA 4.1). By incorporating an "Option to not store" toggle within this interface, the design grants users User Control and Freedom, ensuring that sensitive personal or medical information is retained solely at the user's explicit request.

The Speech-to-Text (STT) interface addresses the requirements of HTA 3.6, providing a dedicated mode for hearing users to interact with the system. The design features a prominent central microphone button, allowing hearing users to record speech which is then processed and displayed as text for the Deaf or mute user to read. By including a "Tap and hold" or "Tap to start" mechanism, the interface ensures the system is only active when intended, reducing background noise interference and enhancing accuracy. This layout maintains the application's commitment to bidirectional communication, ensuring that the

non-signer has a clear, accessible path to respond within the same visual environment as the signer.
Below are the final low-fidelity prototypes illustrating the Premium Learner Features (HTA 5) and the Calibration Demos (HTA 1.4). At the moment, not much emphasis is given to the learning interface since it is a premium feature and not the main intended purpose of the application. Rather it is an additional feature to provide users with an all-rounded experience.

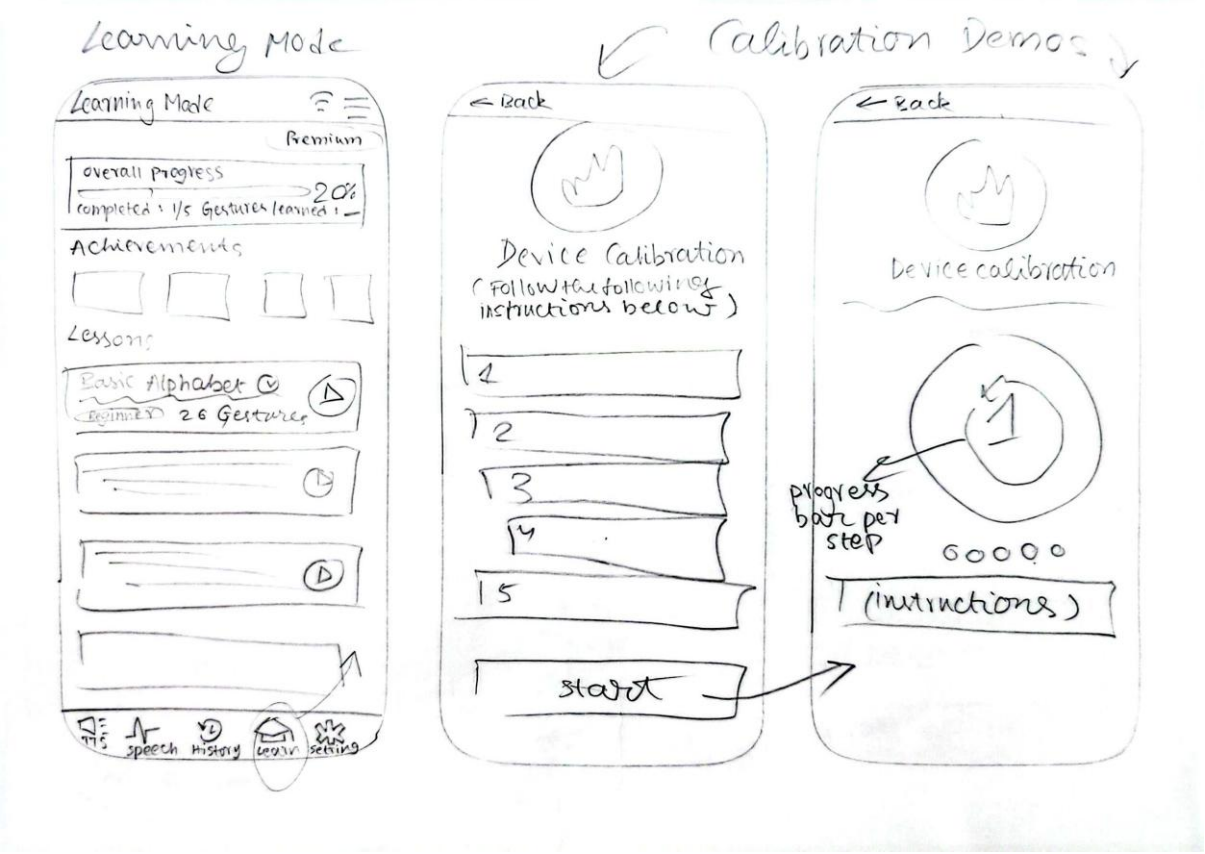


Figure 11: Section 4 of Low-fidelity prototypes

The Learner Mode and Premium Features correspond to HTA 5. This segment of the prototype presents a gamified interface for users aiming to enhance their sign language proficiency through AI-assisted feedback (HTA 5.2). The design includes visual depictions of "Milestones" and badges, which are intended to inspire the user. These premium tasks are deliberately distinguished from the fundamental translation tools to avoid disrupting essential accessibility functions. This distinction guarantees that while the application provides advanced learning and professional applications, the primary objective of immediate communication remains clear and effective.

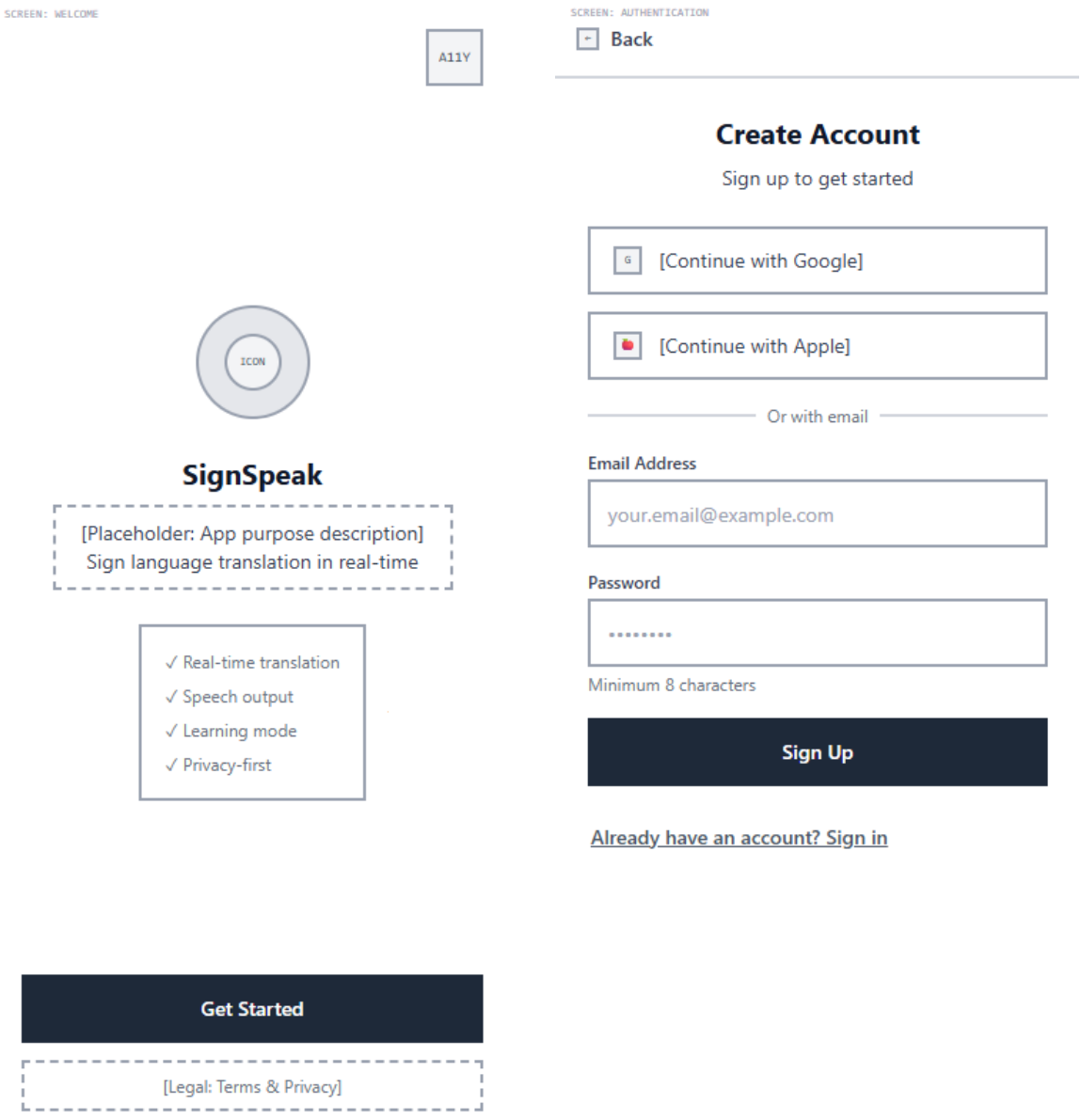
The Calibration Demos represent the practical execution of HTA 1.4 (Pair and calibrate EMG wristband), moving beyond connection to functional accuracy. The prototype utilizes a step-by-step instructional layout where users are prompted to perform specific gestures to synchronize the hardware with the machine learning model. A circular progress bar per step provides immediate feedback on the completion of each gesture, ensuring the user knows exactly how much data is required for a successful calibration. This guided process simplifies

a technically complex task into manageable actions, reducing user frustration and ensuring the translation engine is accurately tuned before primary use begins.

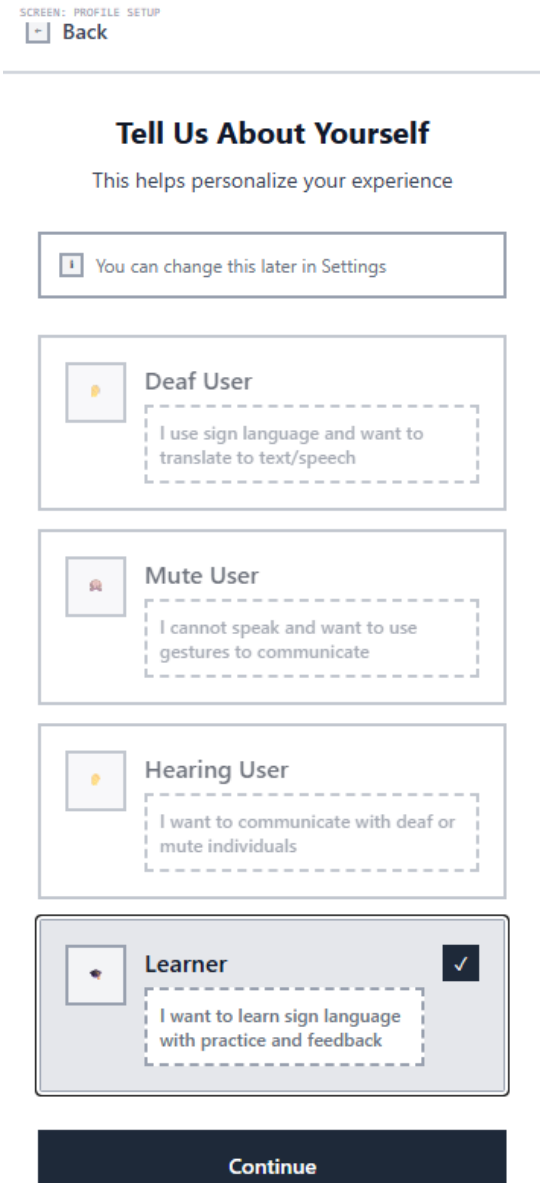
2.5 Digital wireframes

Drawing from the paper prototypes discussed in section 2.4, I developed the digital wireframe for my application using Figma. These wireframes focus on the "skeleton" of the SignSpeak interface, ensuring that the placement of elements supports the user's mental model while adhering to established usability heuristics.

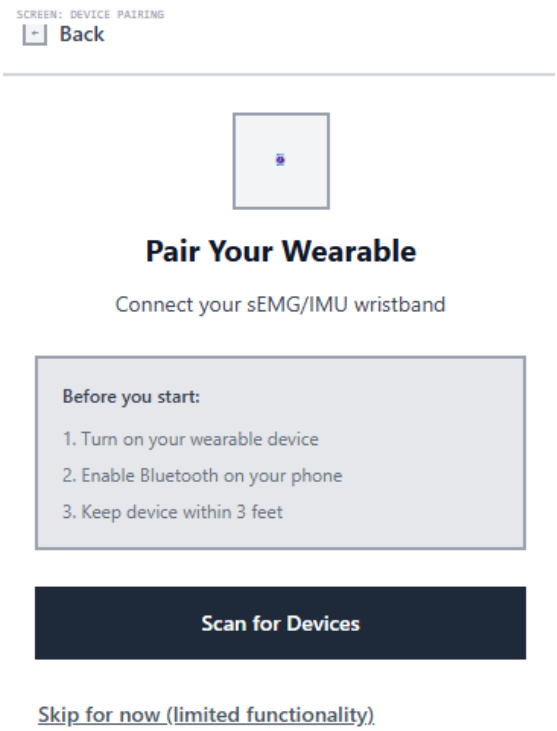
1. Onboarding and Profile Customization (HTA 1.1 – 1.3)

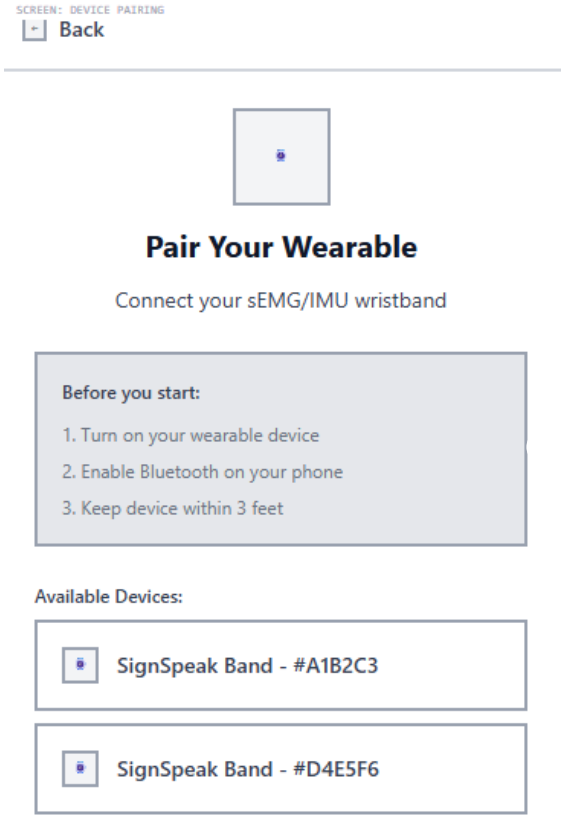



SCREEN: PROFILE SETUP
Back
Tell Us About Yourself
This helps personalize your experience
You can change this later in Settings
Deaf User
I use sign language and want to translate to text/speech
Mute User
I cannot speak and want to use gestures to communicate
Hearing User
I want to communicate with deaf or mute individuals
Learner
I want to learn sign language with practice and feedback
Select a Profile Type


SCREEN: PROFILE SETUP
Back
Tell Us About Yourself
This helps personalize your experience
You can change this later in Settings
Deaf User
I use sign language and want to translate to text/speech
Mute User
I cannot speak and want to use gestures to communicate
Hearing User
I want to communicate with deaf or mute individuals
Learner
I want to learn sign language with practice and feedback
Continue

2. Hardware Synchronization and System Status (HTA 1.4)





SCREEN: CALIBRATION

Back



Device Calibration

Follow instructions for accurate recognition

You'll perform these gestures:

1. **Rest Position**
   Keep hand relaxed at side
2. **Fist Gesture**
   Make a fist
3. **Open Palm**
   Open hand, palm forward
4. **Point Gesture**
   Point index finger up
5. **Wave Motion**
   Wave hand side to side

Tip: Hold each gesture steady for 3 seconds

Start Calibration

SCREEN: CALIBRATION

Back



Device Calibration

Follow instructions for accurate recognition

Step 3 of 5

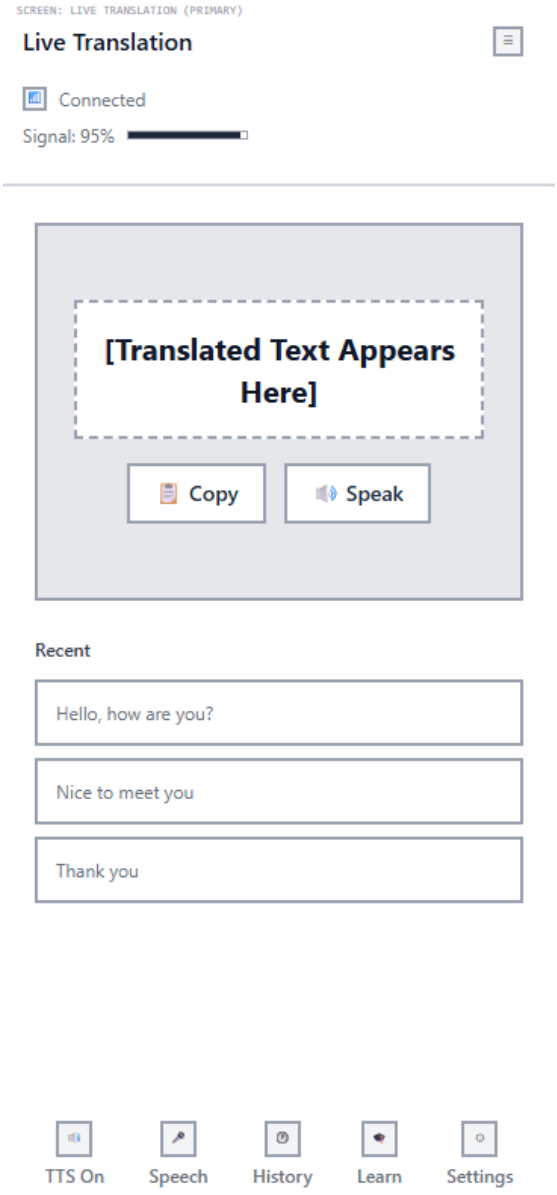
3

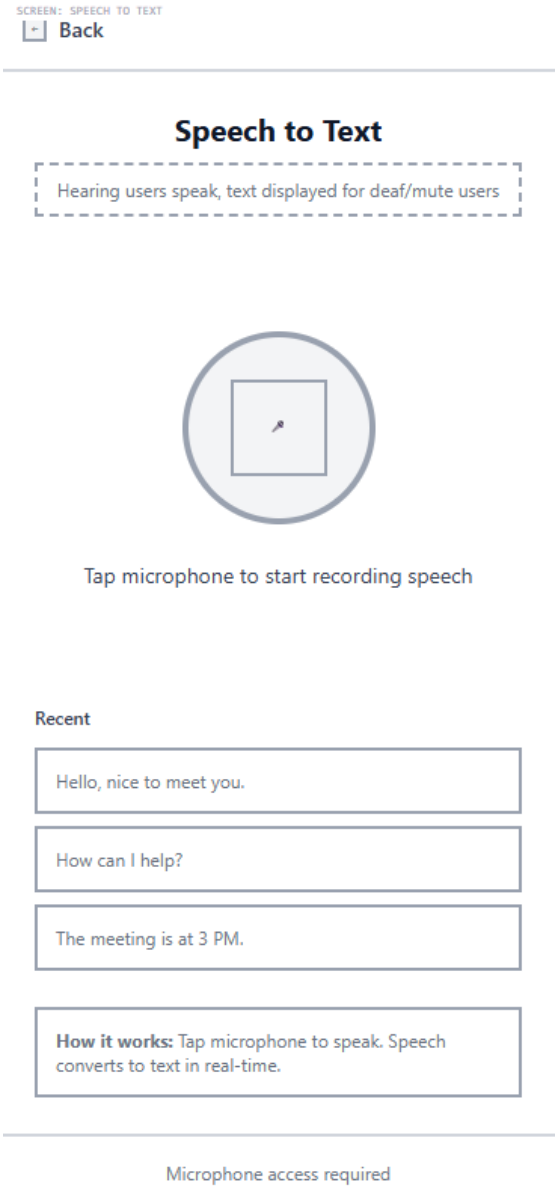
Open Palm

Open hand, palm forward

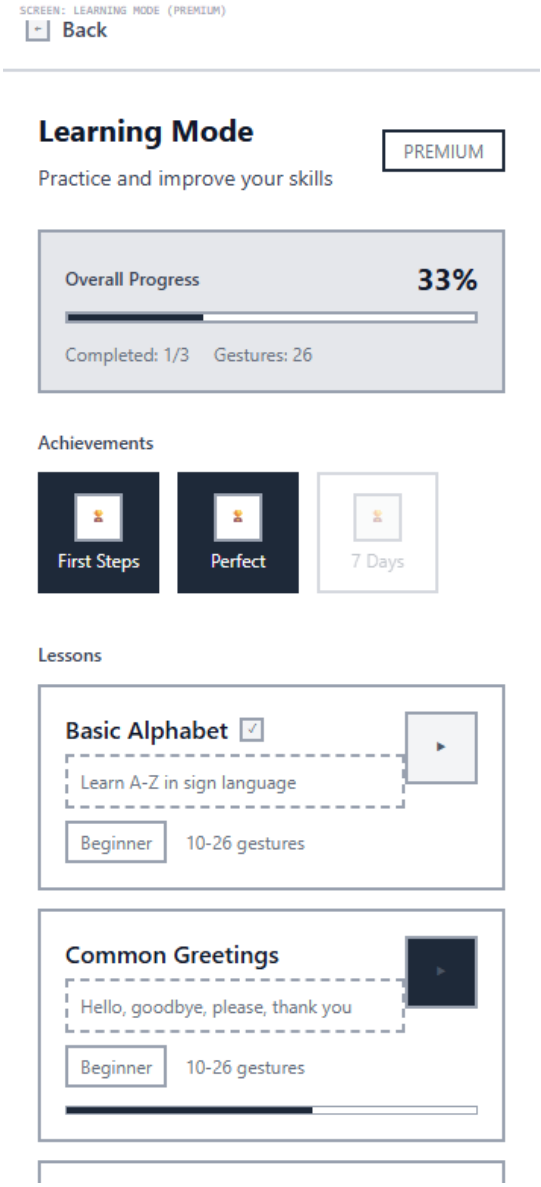
Hold gesture steady...

3. Core Translation and Multimodal Output (HTA 3)



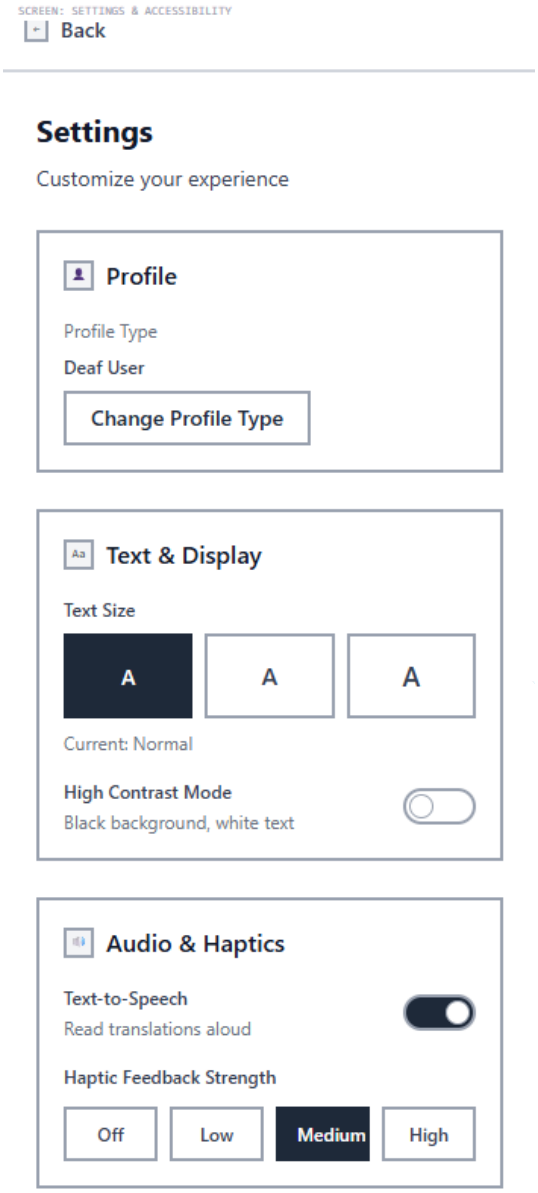


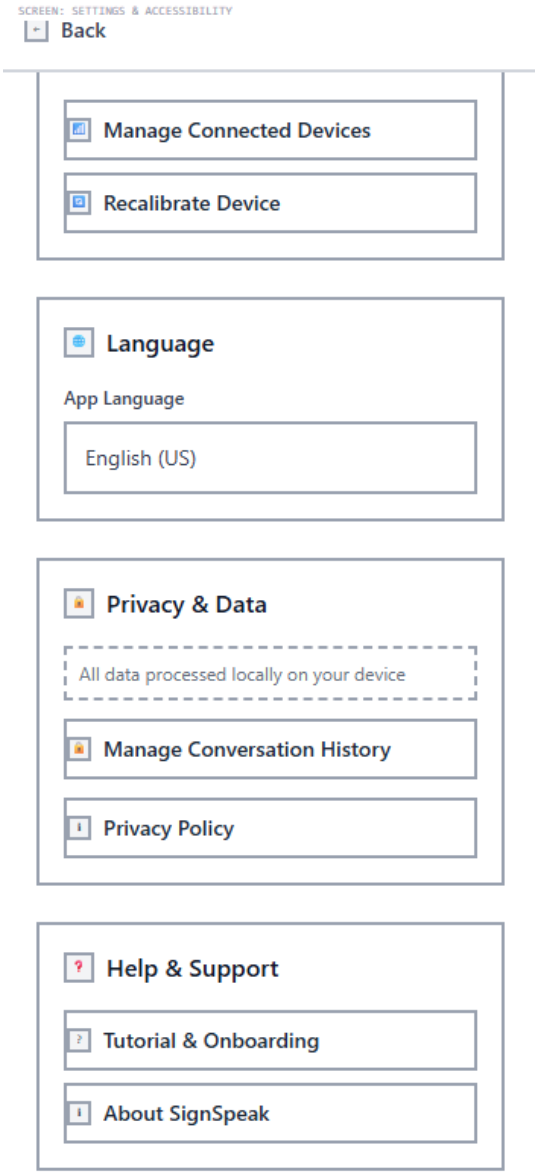
4. Premium Learner Mode (HTA 5)





5. Accessibility Settings and Data Privacy (HTA 2 & 4)





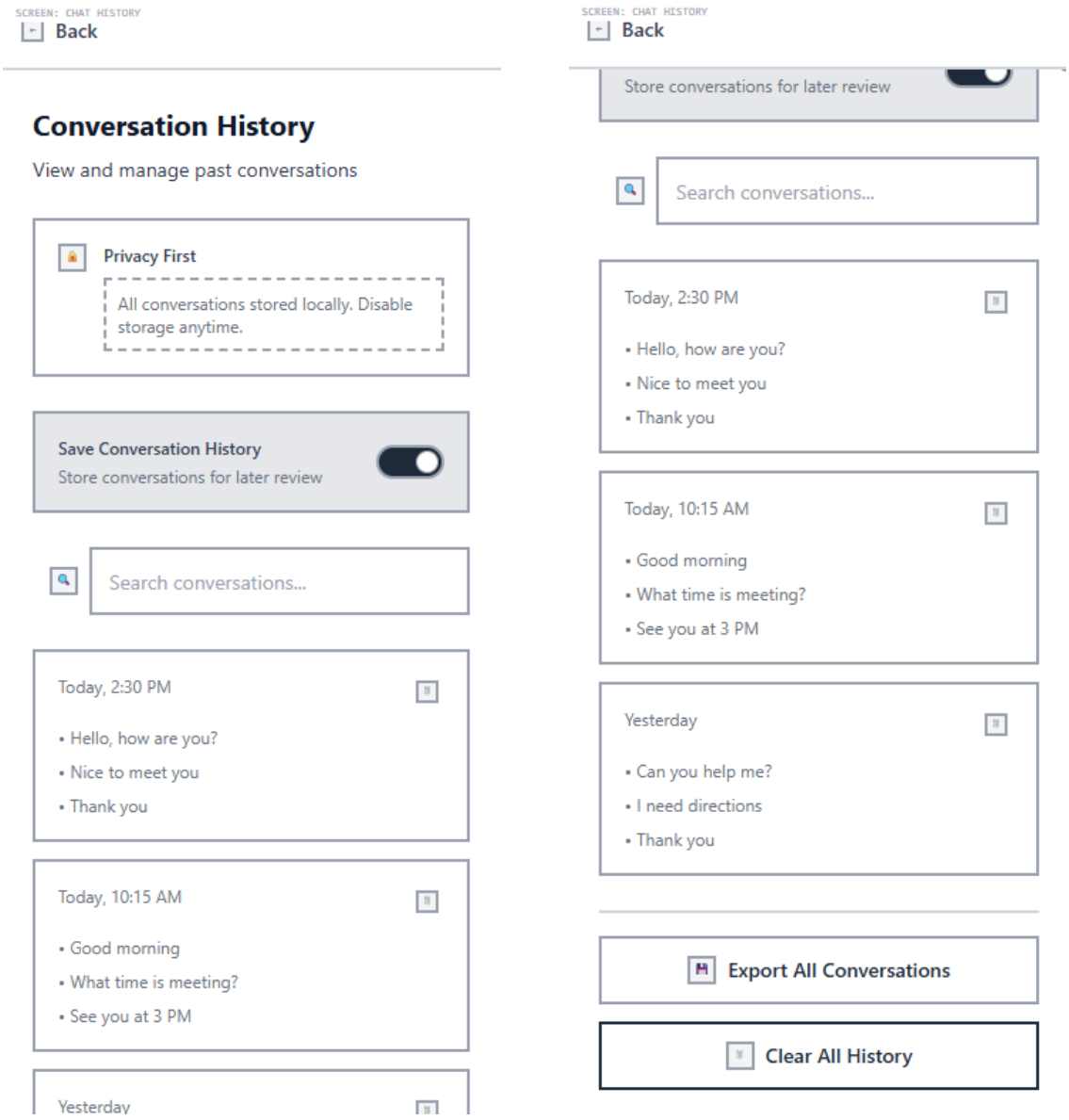


Figure 12: Medium-fidelity prototypes

2.6 High-fidelity prototype (mockup)

Android demo: https://youtu.be/YK01FPR1aSg
IP16 demo: https://youtu.be/rfuTgGKw7RE

The high-fidelity prototype embodies the ultimate integration of functional requirements and visual design principles. Advancing beyond wireframes, this phase establishes a unified brand identity through the application of color theory, sophisticated typography and interactive elements, thereby improving credibility and user engagement, all while rigorously complying with WCAG 2.1 accessibility standards.

A. Brand Identity and Inclusive Onboarding

The visual identity of SignSpeak utilizes a palette of deep navy and vibrant violet, creating a modern, high-tech aesthetic that remains accessible.

- Design Decision: As seen in the figure below of the “Tell us about yourself” section, the onboarding screens use high-contrast text and distinct iconography for the "Deaf," "Mute," and "Hearing" user roles.
- UX Rationale: This follows the Color Contrast Principle, ensuring a ratio of at least 4.5:1 for maximum readability (W3C, 2018). The use of inclusive icons supports Dual Coding Theory, allowing users to identify their profile type through both text and imagery, which significantly reduces the "Interaction Cost" during the initial setup (Paivio, 1971).

B. Real-Time Feedback and Hardware Status

In the high-fidelity Pairing and Device Management screens, micro-interactions are used to communicate system states that are otherwise invisible to the user.

- Design Decision: The pairing process features pulsating animations during the search phase and a clear green "Connected" badge once the sEMG wristband is active.
- UX Rationale: These visual "proxies" for hardware connectivity support Nielsen’s Heuristic of Visibility of System Status (Nielsen, 2020). By providing immediate confirmation of a successful Bluetooth handshake, the design mitigates user anxiety regarding wearable sensor reliability.

C. The Translation Interface

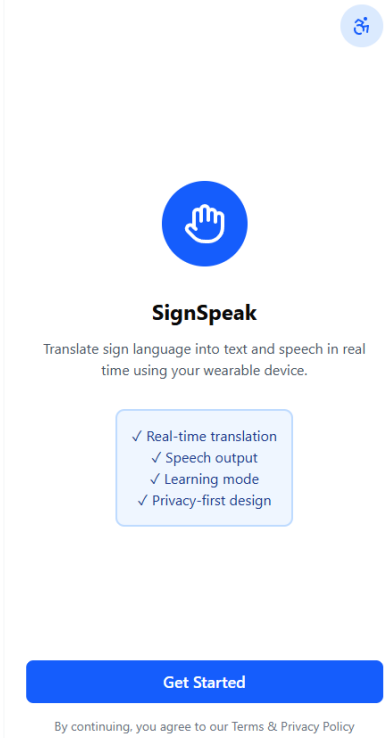
The Main Translation screen is designed to be the functional heart of the app, optimized for real-time, spontaneous conversation.

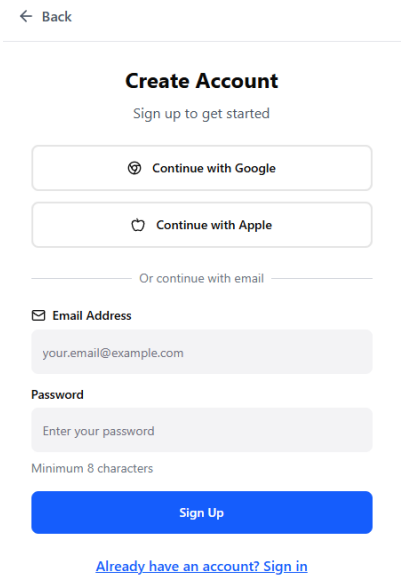
- Design Decision: The interface features a prominent, high-contrast text display area for recognized gestures. Large, tactile "pill-shaped" buttons for Text-to-Speech (HTA 3.5) and Speech-to-Text (HTA 3.6) are positioned at the bottom.
- UX Rationale: Applying Fitts’s Law, these primary targets are sized for easy reachability, which is vital during live interactions where speed is essential (Mackenzie, 1992). Furthermore, a real-time signal waveform is integrated at the top of the screen to provide the user with confidence that their muscle signals are being captured accurately by the AI model.

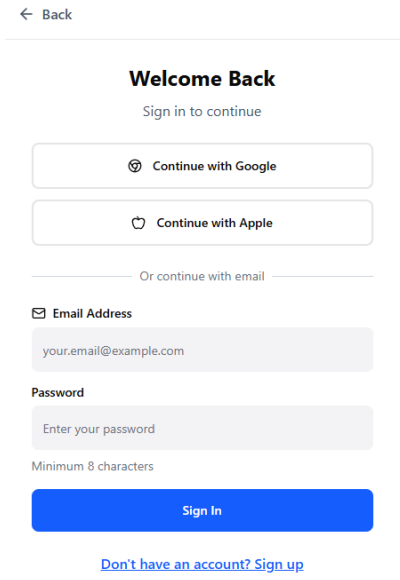
D. Specialized Learning and Accessibility Controls

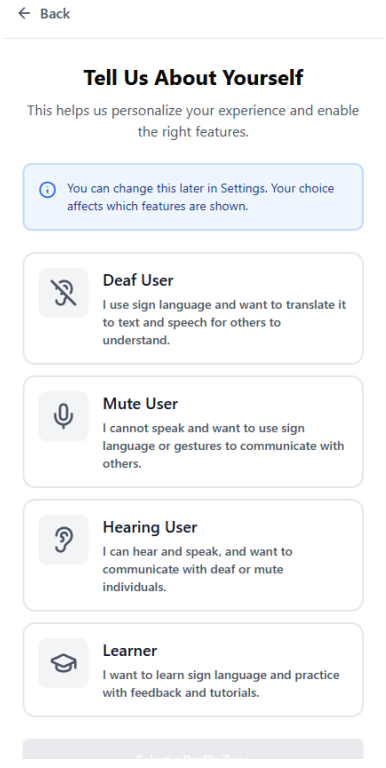
The final screens address the diverse needs of the target audience, from data privacy to educational growth.

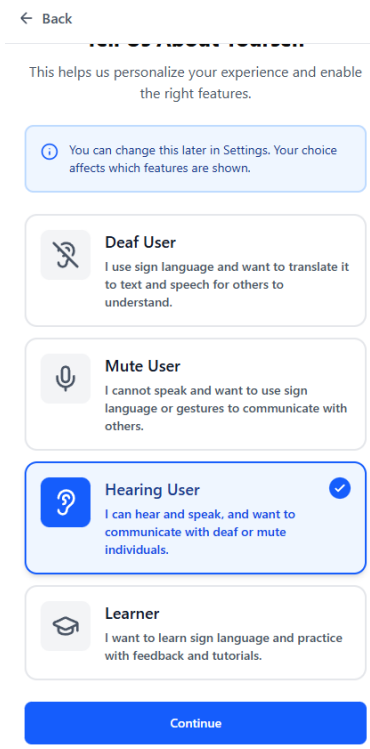
- Design Decision: The Learning Mode (HTA 5) uses gamification elements, such as progress rings and achievement badges, to encourage users to improve their signing skills. In the Settings (HTA 2), a "High Contrast Mode" toggle allows for a complete shift to black-and-white for users with light sensitivity.
- UX Rationale: Providing a dedicated "High Contrast" option is a core tenet of Inclusive Design, ensuring the app serves users with varying visual abilities. The Conversation History (HTA 4) remains minimalist and privacy-focused, with clear "Clear All History" calls-to-action to support User Control and Freedom over their personal data.

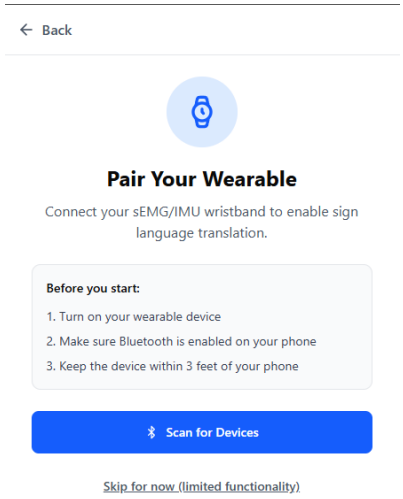

SignSpeak
Translate sign language into text and speech in real time using your wearable device.
✓ Real-time translation
✓ Speech output
✓ Learning mode
✓ Privacy-first design
Get Started
By continuing, you agree to our Terms & Privacy Policy

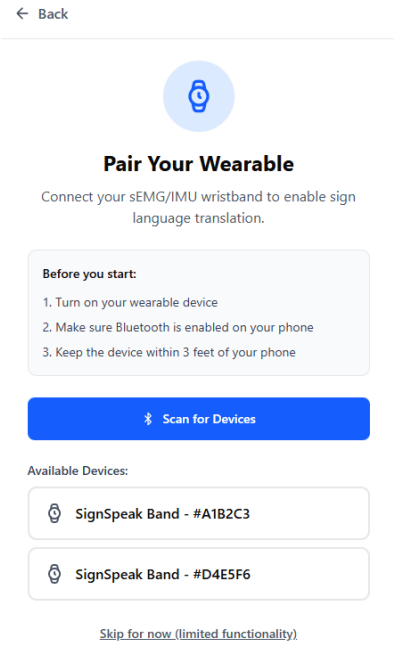

Back
Create Account
Sign up to get started
Continue with Google
Continue with Apple
Or continue with email
Email Address
your.email@example.com
Password
Enter your password
Minimum 8 characters
Sign Up
Already have an account? Sign in

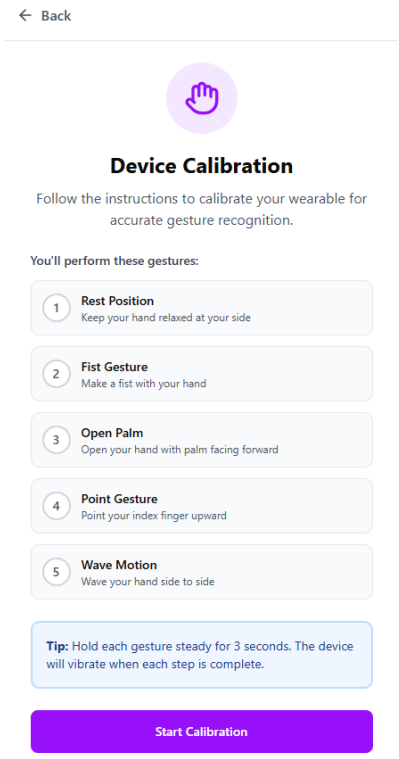

Back
Welcome Back
Sign in to continue
Continue with Google
Continue with Apple
Or continue with email
Email Address
your.email@example.com
Password
Enter your password
Minimum 8 characters
Sign In
Don't have an account? Sign up

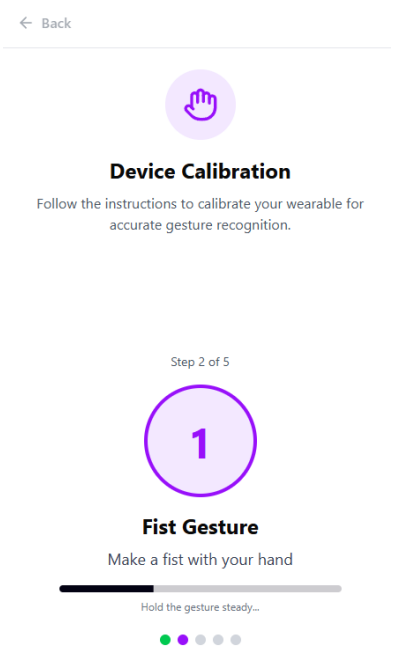

Back
Tell Us About Yourself
This helps us personalize your experience and enable the right features.
You can change this later in Settings. Your choice affects which features are shown.
Deaf User
I use sign language and want to translate it to text and speech for others to understand.
Mute User
I cannot speak and want to use sign language or gestures to communicate with others.
Hearing User
I can hear and speak, and want to communicate with deaf or mute individuals.
Learner
I want to learn sign language and practice with feedback and tutorials.

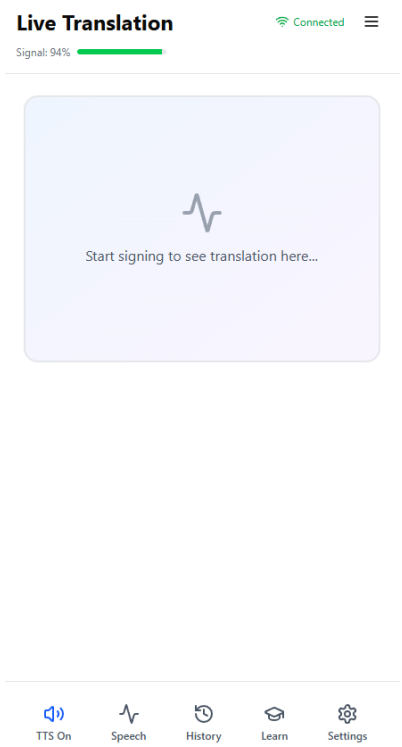

Back
This helps us personalize your experience and enable the right features.
You can change this later in Settings. Your choice affects which features are shown.
Deaf User
I use sign language and want to translate it to text and speech for others to understand.
Mute User
I cannot speak and want to use sign language or gestures to communicate with others.
Hearing User
I can hear and speak, and want to communicate with deaf or mute individuals.
Learner
I want to learn sign language and practice with feedback and tutorials.
Continue

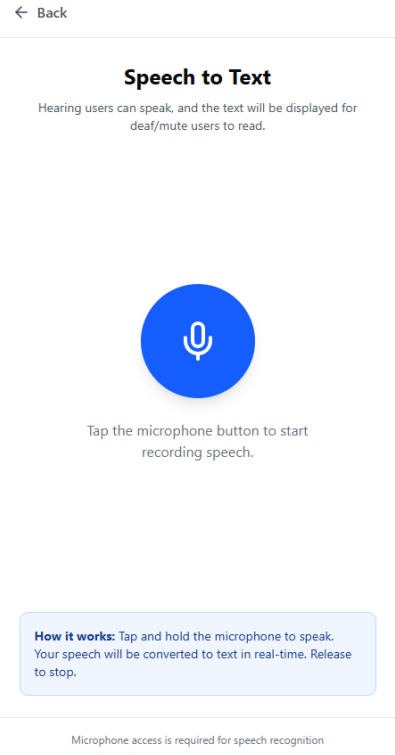

Back
Pair Your Wearable
Connect your sEMG/IMU wristband to enable sign language translation.
Before you start:
1. Turn on your wearable device
2. Make sure Bluetooth is enabled on your phone
3. Keep the device within 3 feet of your phone
Scan for Devices
Skip for now (limited functionality)

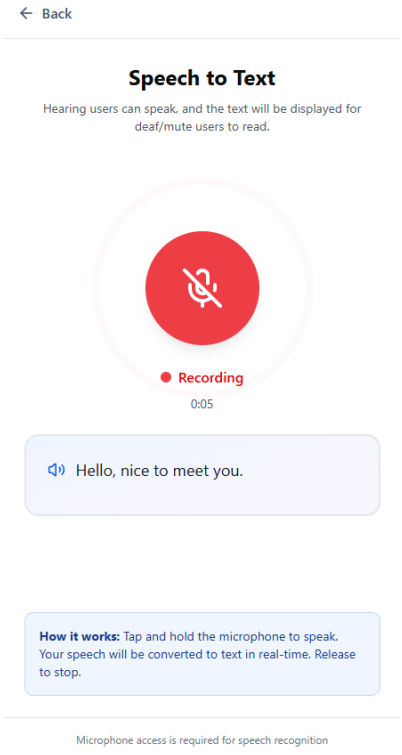

Back
Pair Your Wearable
Connect your sEMG/IMU wristband to enable sign language translation.
Before you start:
1. Turn on your wearable device
2. Make sure Bluetooth is enabled on your phone
3. Keep the device within 3 feet of your phone
Scan for Devices
Available Devices:
SignSpeak Band - #A1B2C3
SignSpeak Band - #D4E5F6
Skip for now (limited functionality)


Back
Device Calibration
Follow the instructions to calibrate your wearable for accurate gesture recognition.
You'll perform these gestures:
1 Rest Position
Keep your hand relaxed at your side
2 Fist Gesture
Make a fist with your hand
3 Open Palm
Open your hand with palm facing forward
4 Point Gesture
Point your index finger upward
5 Wave Motion
Wave your hand side to side
Tip: Hold each gesture steady for 3 seconds. The device will vibrate when each step is complete.
Start Calibration


Back
Device Calibration
Follow the instructions to calibrate your wearable for accurate gesture recognition.
Step 2 of 5
1
Fist Gesture
Make a fist with your hand
Hold the gesture steady...


Live Translation
Connected
Signal: 94%
Start signing to see translation here...
TTS On
Speech
History
Learn
Settings


Back
Speech to Text
Hearing users can speak, and the text will be displayed for deaf/mute users to read.
Tap the microphone button to start recording speech.
How it works: Tap and hold the microphone to speak. Your speech will be converted to text in real-time. Release to stop.
Microphone access is required for speech recognition


Back
Speech to Text
Hearing users can speak, and the text will be displayed for deaf/mute users to read.
Recording
0:05
Hello, nice to meet you.
How it works: Tap and hold the microphone to speak. Your speech will be converted to text in real-time. Release to stop.
Microphone access is required for speech recognition

Back

Conversation History

View and manage your past conversations

Privacy First

All conversations are stored locally on your device. You can disable storage at any time.

Save Conversation History Store conversations for later review

Search conversations...

Today, 2:30 PM

- Hello, how are you?
- Nice to meet you
- Thank you for your help

Today, 10:15 AM

- Good morning
- What time is the meeting?
- See you at 3 PM

Yesterday

- Can you help me?

Back

Learning Mode

Premium

Practice and improve your sign language skills

Overall Progress 20%

Completed: **1/5** Gestures Learned: **26**

Achievements

First Steps | Perfect Score | 7 Day Streak

Lessons

Basic Alphabet

Learn A-Z in sign language

Beginner 26 gestures

Common Greetings

Hello, goodbye, please, thank you

Beginner 10 gestures • 60% complete

Numbers 1-10

Practice counting in sign language

Beginner 10 gestures

Back

Today, 2:30 PM

- Hello, how are you?
- Nice to meet you
- Thank you for your help

Today, 10:15 AM

- Good morning
- What time is the meeting?
- See you at 3 PM

Yesterday

- Can you help me?
- I need directions
- Thank you so much

Dec 12, 2025

- How was your day?
- That sounds great
- Talk to you later

Export All Conversations

Clear All History

Back

Settings

Customize your experience and manage your account

Profile

Profile Type

Hearing User

Change Profile Type

Text & Display

Text Size

A | A | A

Current: Normal

High Contrast Mode
Black background with white text

Audio & Haptics

Text-to-Speech Read translations aloud

Haptic Feedback Strength

Off | Low | Medium | High

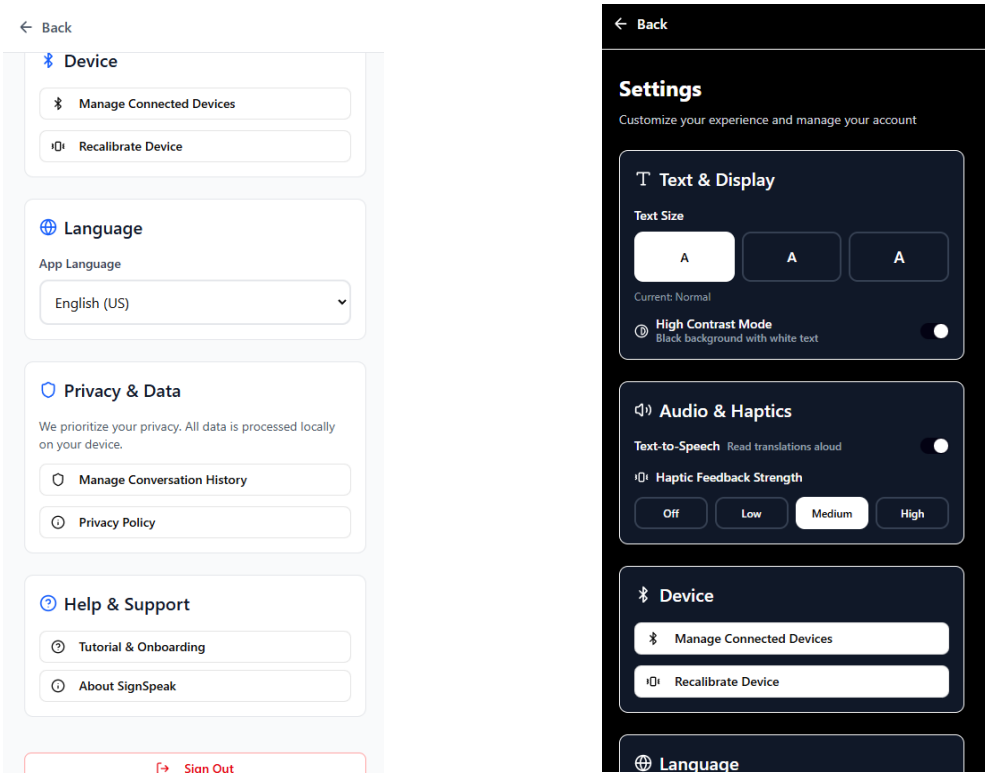


Figure 13: High-fidelity prototypes

3 Usability Testing

Usability testing is a method of evaluating a product by requiring representative users to perform specific tasks while observers collect data to identify usability issues and gather feedback (Usability.gov, n.d.).

3.1 Protocol

For SignSpeak, the usability test was designed to explore the effectiveness of the mobile interface as a standalone platform. Since the physical sEMG/IMU wristband is currently under development, this test focused on the software's ability to guide users through the setup and translation process. Five users participated, as this is the optimal size for identifying core usability issues (Nielsen, 2000).

3.1.1 Methodology:

The test was carried out using a hybrid moderated approach, combining in-person sessions with remote digital evaluations. For the digital interaction, I utilized the Figma Mirror environment, which allowed participants to interact with the high-fidelity prototype as a standalone mobile application on a smartphone.

The choice of moderated testing was critical for this project because the physical sEMG/IMU hardware is currently in the development phase. Moderation allowed me to:

- Provide necessary context regarding the "missing" wearable gadget and simulate the wristband's role in the system.
- Observe real-time user behaviors, specifically focusing on how easily they navigated the app logic without physical hardware feedback.

- Offer immediate support if the prototype's non-functional or static elements caused confusion.

The session was designed as a structured path: participants first completed the interactive tasks on the prototype, followed by a Google Form to capture quantitative SUS scores and a semi-structured interview for qualitative insights.

3.1.2 Sessions:

A total of five sessions were conducted between December 29 and January 12, lasting approximately 30 minutes each. To ensure a diverse range of feedback and environmental testing, I utilized two different settings:

- In-Person Testing: Three sessions were conducted on-site with classmates at the UIT Library. These face-to-face sessions allowed me to observe natural hand gestures and ergonomic comfort while users navigated the translation interface. The chosen setting simulating a real-world use case for the app.
- Remote Testing: Two sessions were conducted online via Google Meet. During these sessions, I presented the testing link and participants shared their screens while interacting with the Figma prototype. This allowed me to evaluate the digital navigation flow and capture feedback from users in a remote, home-based environment.

Before each session began, I welcomed the participants and briefed them on the project's goal of bridging the communication gap. I explicitly stated that the focus of this test was on the app's interface logic and navigation and not the hardware's sensor accuracy. After completing the tasks using the "Think Aloud" method, I provided the participants with a link to a Google Form to submit their satisfaction ratings and SUS responses.

3.2 Data Collection Methods

To produce a comprehensive analysis of the SignSpeak prototype, I collected both qualitative and quantitative data.

3.2.1 Think Aloud Method (Qualitative)

I employed the Think Aloud method which involves participants verbalizing their thoughts, feelings and intentions while interacting with the interface. According to the Nielsen Norman Group (2012), this is the most effective way to understand a user's mental model. It was particularly useful in the UIT Library sessions to identify if my classmates found the navigation intuitive or if they were confused by the simulated hardware pairing process.

3.2.2 System Usability Scale - SUS (Quantitative)

Following the tasks, I provided a Google Form containing the System Usability Scale (SUS). SUS is a "quick and dirty" but highly reliable tool for measuring usability, consisting of a 10-item questionnaire (Brooke, 1996). I chose this method because it is an industry standard that allows me to convert subjective user feelings into a numerical score, making it easier to identify if the app meets the "average" usability threshold of 68.

3.2.3 Semi-Structured Interviews (Qualitative)

The final stage of my data collection was a semi-structured interview. Unlike a typically rigid survey, this allowed for open-ended conversation where I could follow up on specific "pain points" I observed during the session. This was crucial for the remote Google Meet sessions,

where I needed to probe deeper into how the participants felt about using a sign-language app without the physical gadget present.

3.3 Participants

A group of five participants with varying levels of sign language experience, technical expertise, and ages was recruited to participate in the usability testing sessions. Recruitment was conducted via email and direct outreach within the UIT class-mates and personal networks. To protect participant confidentiality, precise identities and exact ages have been excluded from this report; however, their signed consent forms are available in Appendix section.

The group represents a diverse demographic split, allowing for feedback from both the local student population in a physical setting and remote users in a digital environment.

Participant	Age Group	Type of User	Familiarity with Sign Language
1	18-24	Hearing user (Non-signer)	Not familiar at all
2	18-24	Sign Language Learner	Basic Knowledge
3	18-24	Hearing user (Non-signer)	Not familiar at all
4	18-24	Sign Language Learner	Not familiar at all
5	25-34	Hearing user (Non-signer)	Not familiar at all

Table 2: Participants data

TESTING INFORMATION SHEET

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Duration: December 29,2025 - January 5,2026

Researcher: Sombatla Truc Deydeepya Lakshmi

Institution: Vietnam National University of Information Technology (UIT)

You are invited to participate in a research study evaluating the user interface of **SignSpeak**, a mobile application designed to translate sign language into text and speech. Before you decide, it is important to understand why the research is being done and what it will involve. Please take the time to read the following information.

What is the purpose of this research? This research aims to evaluate the navigation logic and user experience of the SignSpeak app. While the final product will eventually include a wearable sensor (sEMG/IMU wristband), this test focuses exclusively on how intuitive the mobile software is for onboarding, calibration, and translation tasks.

Why have I been invited? You have been invited because you are either a student at UIT or a potential user who can provide feedback on the app's logic. We are looking for 5 participants to help identify friction points in the design.

What does participation involve? If you agree to take part, you will be asked to:

- Complete 3-4 specific tasks using the **Figma** prototype on a smartphone (in-person) or via screen-share (remote).
- Use the "**Think Aloud**" method (verbalize your thoughts as you navigate).
- Complete a 10-item **System Usability Scale (SUS)** survey on a Google Form.
- Participate in a brief 5-minute exit interview. The entire session will last approximately **30 minutes**.

Is this compulsory to take part in? Your participation is entirely voluntary. You can withdraw at any time during the session without giving a reason and without any penalty.

How will my data be used? All data collected (notes, SUS scores, and interview responses) will be **anonymized**. You will be referred to as "Participant X" (e.g., P1, P2) in the final report. No personal identification will be shared.

Who can I contact for further information? If you have questions, you can contact the researcher at: ddpyawork@gmail.com

The following participants were handed the following information sheet and briefed on what they were participating in and made to sign the consent form next to the information sheet.

CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: _____

Participant Age (if under 18 years of age) : _____

Date: _____

Signature: _____

3.4 Results

3.4.1 Quantitative Results

Closed-ended questions & Demographics: These questions were designed to gather information about the users' backgrounds and expectations to contextualize their performance.

- **Demographics:** 80% (4 out of 5) of the respondents are aged 18-24, with one participant in the 25-34 range. 80% of the sample identified as Hearing Users, providing a "fresh eyes" perspective on the app's onboarding clarity.
- **Sign Language Familiarity:** The group showed low baseline familiarity; 80% were "Not familiar at all" with sign language. This was beneficial for testing the "Calibration" instructions, as it ensured the UI was teaching the gestures effectively to novices.
- **Technical Experience:** 80% of participants have used assistive or translation technology occasionally or regularly before, indicating a tech-savvy user base.
- **User Expectations:** When asked about the biggest challenge for this technology, respondents were divided between **Speed/Delay in output (60%)**, **Comfort (20%)**, and **Trust in the system (20%)**.

Task Performance & Design Metrics: The following metrics evaluate the efficiency of the interface based on a 1-5 Likert scale:

- **Initial Clarity:** Participants gave a **4.8/5.0** rating for the clarity of the app's purpose upon opening. 100% of participants felt confident about what to do next after seeing the home screen.
- **Navigation Success:** Finding the option to connect the wearable device received a **4.4/5.0**. One participant (P4) rated this a **3/5**, noting a slight delay in locating the Bluetooth pairing button.
- **Calibration Logic:** Understanding the instructions for the "Fist" and "Wave" gestures was rated highly at **4.8/5.0**, suggesting the visual guides are effective.
- **Accessibility:** 80% of users felt the app was definitively designed with accessibility in mind. The text size and contrast received a high readability score of **4.6/5.0**.

System Usability Scale (SUS): The system received an overall score of **89.0**, placing it in the "Excellent" bracket. This indicates a high level of comfort and ease of use. However, a minor concern was raised by P4 (the lowest score of 72.5), who felt "uncomfortable" with the calibration process, which directly impacted their perception of the system's ease of use.

Participant	SUS Score Calculation	SUS Score	Grade	Adjective Rating
P1	(18+16)*2.5 = 34*2.5	85.0	A	Excellent
P2	(18+20)*2.5 = 38*2.5	95.0	A+	Best Imaginable
P3	(18+19)*2.5 = 37*2.5	92.5	A+	Best Imaginable
P4	(14+15)*2.5 = 29*2.5	72.5	B	Good
P5	(20+20)*2.5= 40*2.5	100.0	A+	Best Imaginable
AVG		89.0	A	Excellent

Table 2: SUS calculation and rating

3.4.2 Qualitative Results (Think Aloud & Interviews)

Think-Aloud Analysis: Overall, all participants accurately identified the core objectives of the app. Their verbal feedback during the tasks provided the following insights:

- **Task: Device Calibration:**
    - P2 noted: *"It was easy to use and quite intuitive,"* during the gesture setup.
    - P4 expressed a "think-aloud" friction point, stating they felt *"uncomfortable"* during the calibration. This indicates that without the physical weight of the wristband, the software-only simulation felt abstract for some users.
- **Task: Translation Mode:**
    - Most participants navigated the translation screen with a "confident tone." P5 remarked that *"Nothing"* was confusing about the navigation and that the interface was *"visually clear and uncluttered."*

Open-Ended Feedback & Interviews: The app was generally well-received, particularly for its specialized focus. However, advanced users suggested more "two-way" depth.

- Initial Impressions: Users described the concept as "Interesting" and "Good." Before starting, 100% of users understood they were testing an app meant to bridge a communication gap.
- Likes and Dislikes: The most valued features were Real-time translation and Translation History. The "Least Liked" aspect was the manual nature of the calibration; users expressed that they would prefer the band to "auto-detect" gestures faster.
- Feature Suggestions:
    - A significant suggestion came from P4, who proposed “Since u already have sign detection, I want to add the reverse capability: Text-to-Sign Translation. This feature takes paragraphs of text and visualizes them using a 3D hand model. This effectively turns our application into a complete two-way translation tool for the deaf community”. They also suggested “that it would be cooler if the output would be coherent sentences and not just words”.
    - P2 noted that while the prototype was good, the experience *"would be better if the band was available too,"* highlighting that the software is highly dependent on the final hardware feel.

4 Analysis/Discussion

The usability testing of the SignSpeak prototype yielded highly positive quantitative results, with a System Usability Scale (SUS) score of 89.0, placing the application in the top 10% of usable software products. However, the qualitative insights revealed critical areas where the interface logic and feature set can be improved to better serve the Deaf and Mute community.

4.1 Recommendations for Improvement

Based on the feedback from the five participants and the identified friction points, the following improvements are recommended:

- Implement Bidirectional Translation (Text-to-Sign): The most significant feedback (specifically from P4) was the desire for a "two-way" tool. Currently, SignSpeak excels at Sign-to-Text/Speech. Future iterations should include a feature where a hearing user can type or speak into the app, which is then translated into sign language via a 3D animated avatar. This would create a truly balanced communication bridge.

- Enhance Calibration Feedback: While the calibration instructions were rated 4.8/5.0, users felt "uncomfortable" or uncertain during the actual process.
- Optimize Output Toggling: The lower rating for switching from text to speech (4.0/5.0) suggests that the toggle is not as responsive as users expect. The UI should be updated to provide an immediate "Ready" status indicator when Speech-to-Text is enabled, ensuring the user knows the system is active before they begin signing.
- Persistent Device Status: To increase "Trust in the system" (a concern noted by 20% of users), a persistent connection status icon should be added to the main translation screen. This would allow users to see the battery and signal strength of their wearable wristband without having to navigate into the settings menu.

4.2 Alternative Testing Techniques

While the hybrid moderated approach and SUS provided a comprehensive overview, other techniques could have offered different dimensions of insight:

- Heuristic Evaluation: A process where 3–5 experts evaluate the interface against recognized usability principles (Nielsen's 10 Heuristics).
    - Strengths: It is fast and can identify major "usability catastrophes" before real users ever see the app (Nielsen, 1994).
    - Weaknesses: It lacks the "human" emotional feedback and may miss issues that only a member of the Deaf community would notice.
- Eye Tracking Analysis: Using specialized hardware to track exactly where a user's gaze falls on the screen.
    - Strengths: This would be particularly valuable for SignSpeak to see if users are looking at the translated text or the gesture guide during calibration.
    - Weaknesses: It is highly expensive, requires a lab setting, and cannot easily be done remotely via Google Meet (Duchowski, 2017).
- Unmoderated Remote Testing (Scale): Using tools like Useberry or Maze to send the prototype to hundreds of users who complete tasks without a researcher present.
    - Strengths: Provides a much larger sample size, making the SUS score more statistically significant.
    - Weaknesses: The researcher cannot ask "Why?" when a user gets stuck, losing the rich qualitative data found in a "Think Aloud" session.

5 References

1. Usability.gov. (n.d.). *Personas*.
2. Interaction Design Foundation. (n.d.). *User Scenarios*.
3. Stanton, N. A., Salmon, P. M., Walker, G. H., Baber, C., & Jenkins, D. P. (2005). *Human Factors Methods: A Practical Guide for Engineering and Design*. Ashgate.
4. Brooke, J. (1996) 'SUS: A "quick and dirty" usability scale', in Jordan, P. W., Thomas, B., McClelland, I. L. and Weerdmeester, B. (eds.) *Usability Evaluation in Industry*. London: Taylor & Francis, pp. 189–194.
5. Duchowski, A. T. (2017) *Eye Tracking Methodology: Theory and Practice*. 3rd edn. Cham: Springer.
6. Nielsen, J. (1994) 'Heuristic evaluation', in Nielsen, J. and Mack, R. L. (eds.) *Usability Inspection Methods*. New York: John Wiley & Sons, pp. 25–62.
7. Sauro, J. and Lewis, J. R. (2016) *Quantifying the User Experience: Practical Statistics for User Research*. 2nd edn. Cambridge, MA: Morgan Kaufmann. *(Note: Use this if you discuss the validity of your 89.0 SUS score)*.

6 Appendices

Consent Forms:

CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES
I agree to take part in the above study.	YES

Participant Name [redacted]

Participant Age (if under 18 years of age): 21

Date: January 3rd , 2026

Signature:



CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

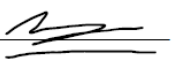
Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : 22

Date: 30 / 12/ 2025

Signature: _____



CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : 20

Date: 12/01/2026

Signature:

CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : 20

Date: 01/01/2026

Signature: Satwik Biswas

CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

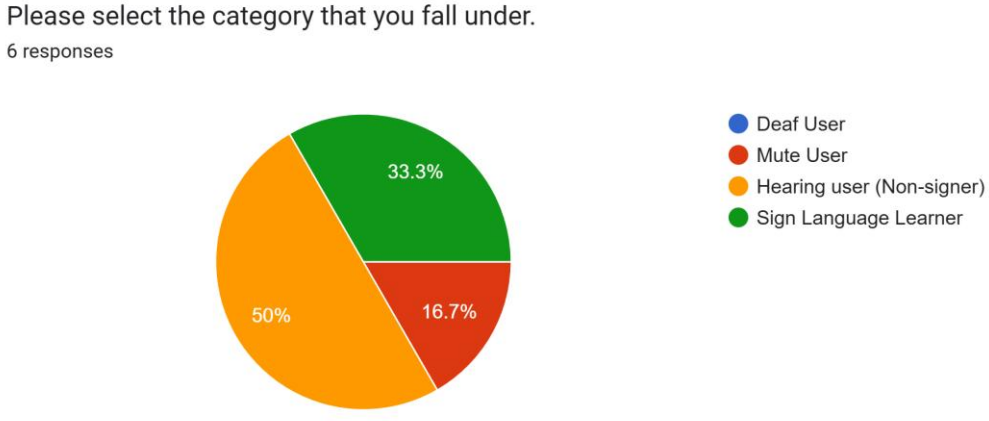
Participant Age (if under 18 years of age) : 27

Date: 3/01/2026

Signature:

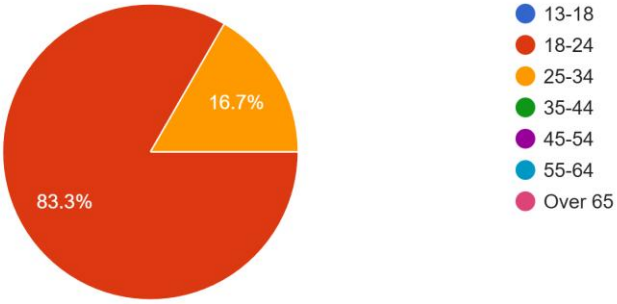


Quiz results:



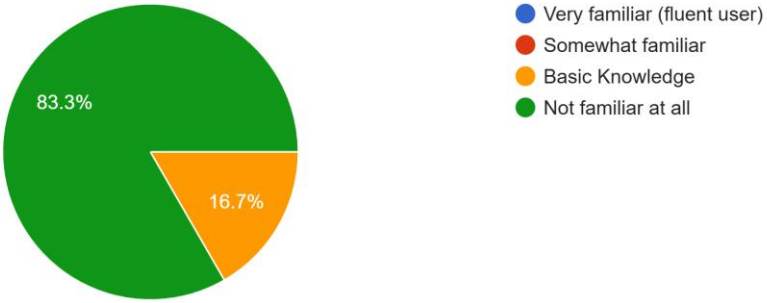
Age

6 responses



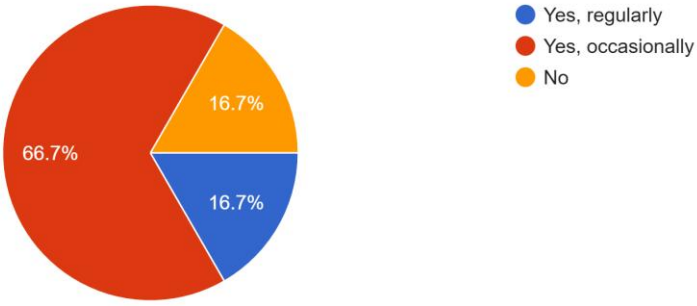
How familiar are you with sign language?

6 responses



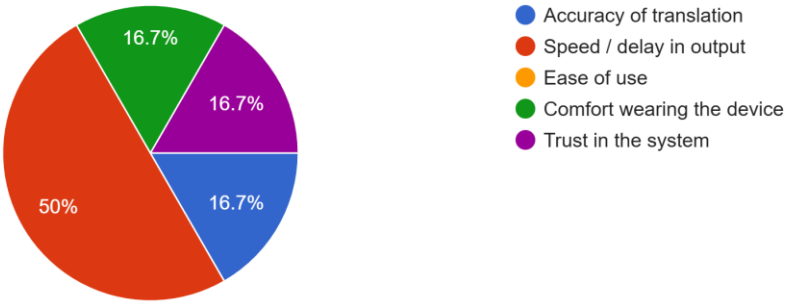
Have you ever used assistive or translation technology before?

6 responses



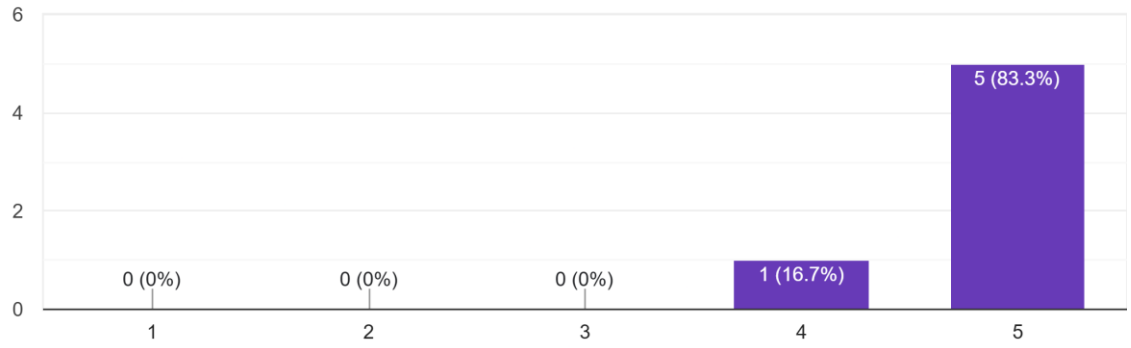
In what context would you most likely use a sign language translation app? (e.g., daily communication, education, public services, emerge...enge when using a sign language translation app?

6 responses



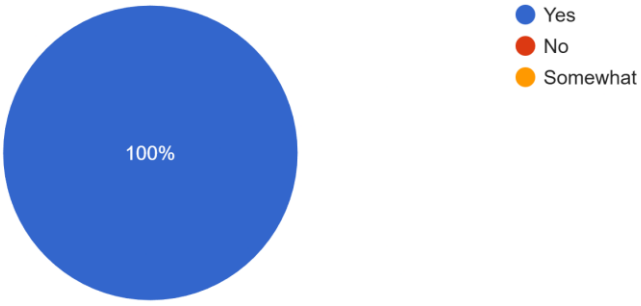
How clear was the purpose of the app when you first opened it? [1 (Not clear at all) – 5 (Very clear)]

6 responses



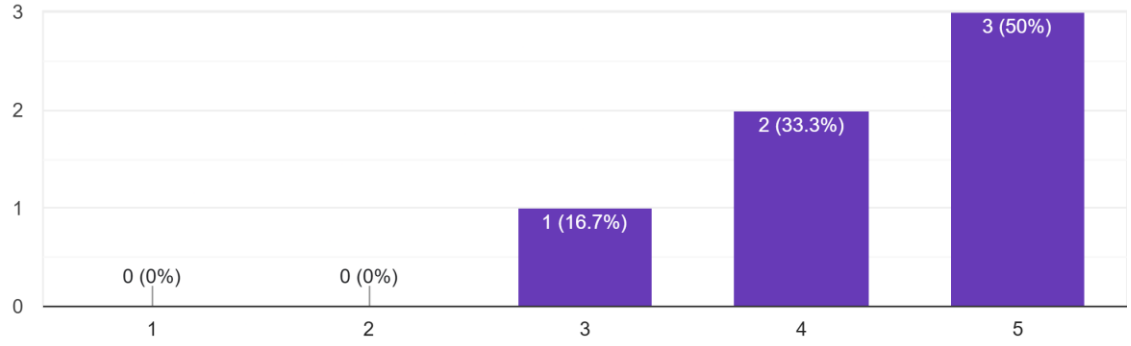
Did you feel confident about what to do next after seeing the home screen?

6 responses



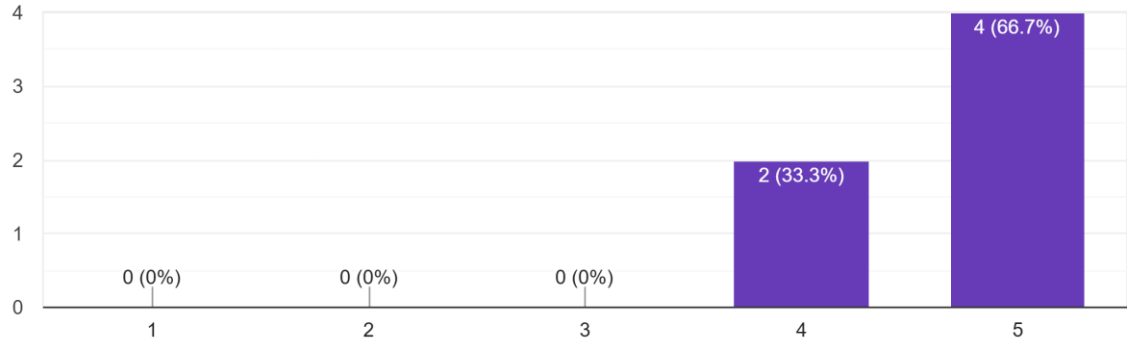
How easy was it to find and use the option to connect the wearable device? [1 (Very difficult) – 5 (Very easy)]

6 responses



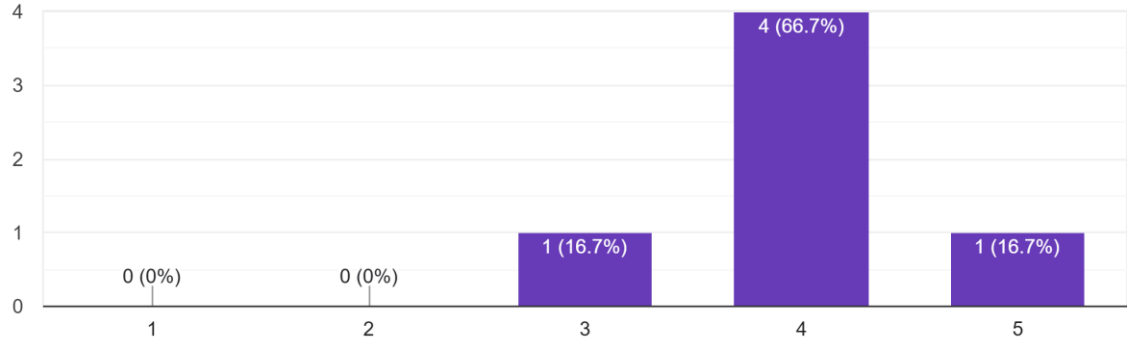
How easy was it to understand the calibration instructions? [1 (Very difficult) – 5 (Very easy)]

6 responses



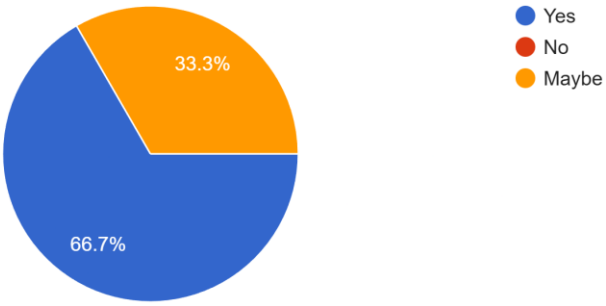
How easy was it to switch from text output to speech output? [1 (Very difficult) – 5 (Very easy)]

6 responses



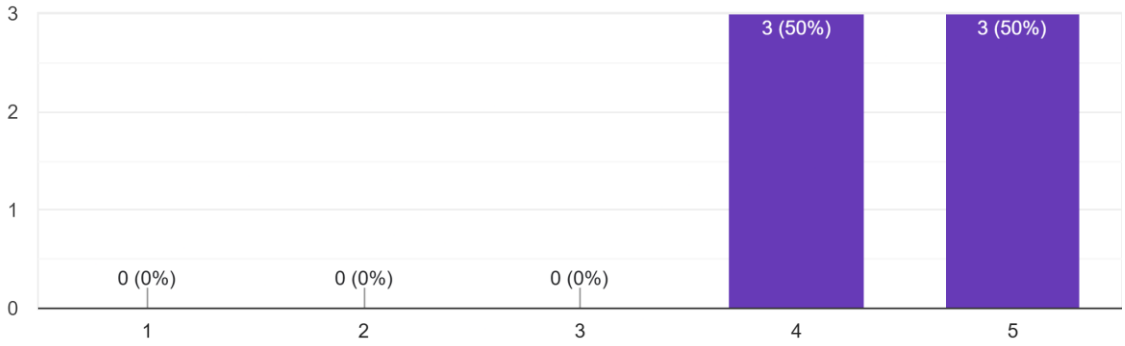
Did you feel the app was designed with accessibility in mind?

6 responses



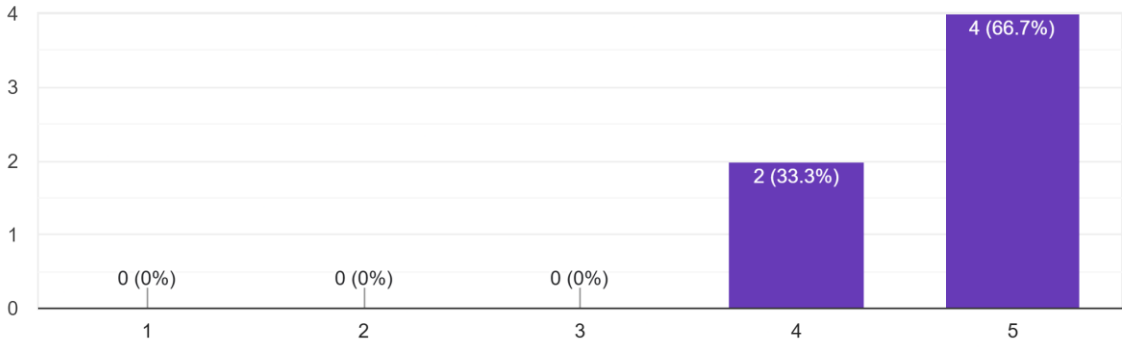
How readable was the text size and contrast? [1 (Very difficult) – 5 (Very easy)]

6 responses



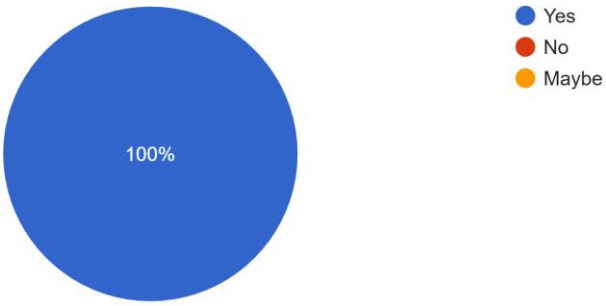
How clear and readable was the translated text on screen? [1 (Very difficult) – 5 (Very easy)]

6 responses



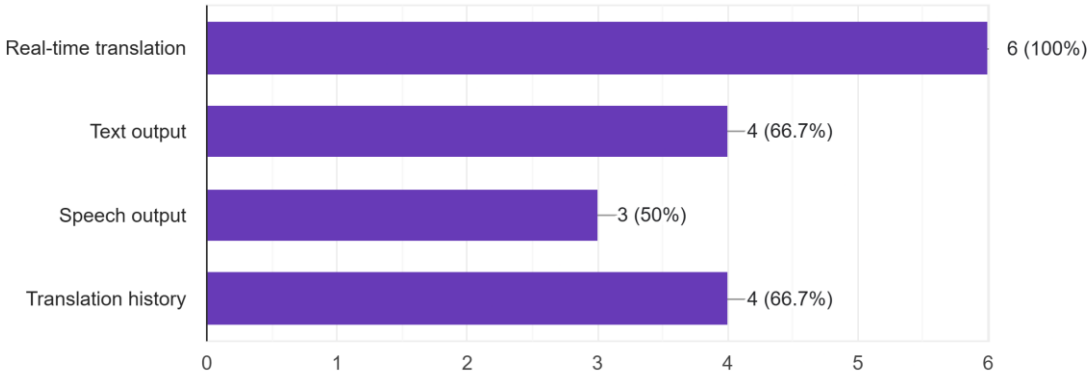
Did the interface feel visually clear and uncluttered?

6 responses



Which features did you find most valuable? (Select all that apply)

6 responses



What additional feature would most improve your experience?

2 responses

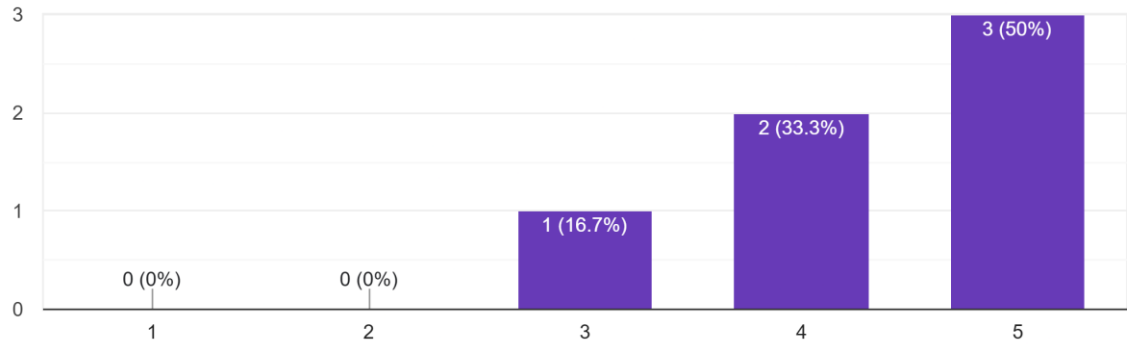
Currently unsure :/

I want to input a full story or text passage and see it translated into a sequence of 3D hand signs. This allows me to see how a continuous narrative looks in sign language, rather than just isolated words.

SUS:

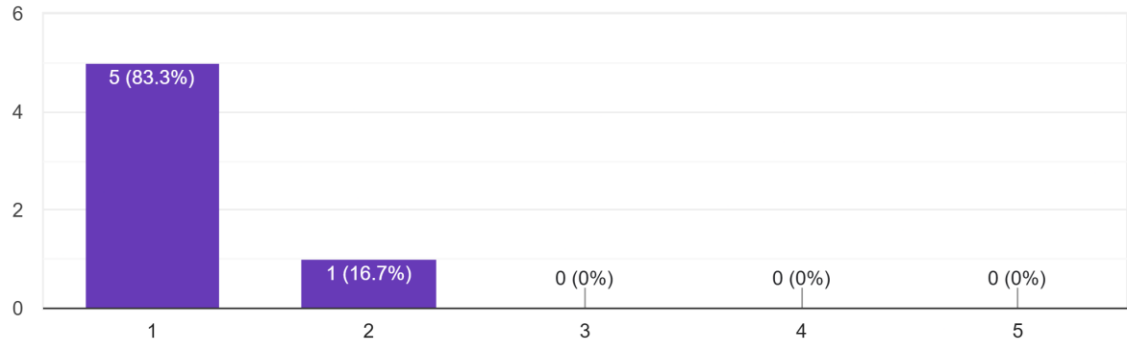
I think that I would like to use SignSpeak frequently.

6 responses



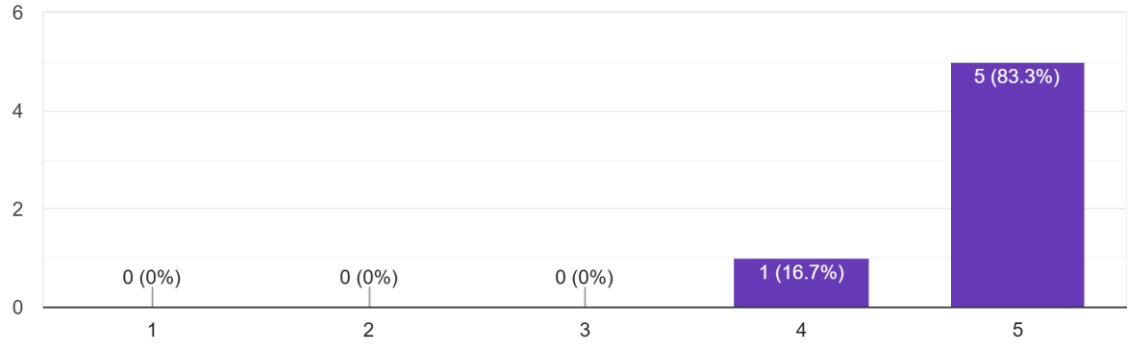
I found the app unnecessarily complex.

6 responses



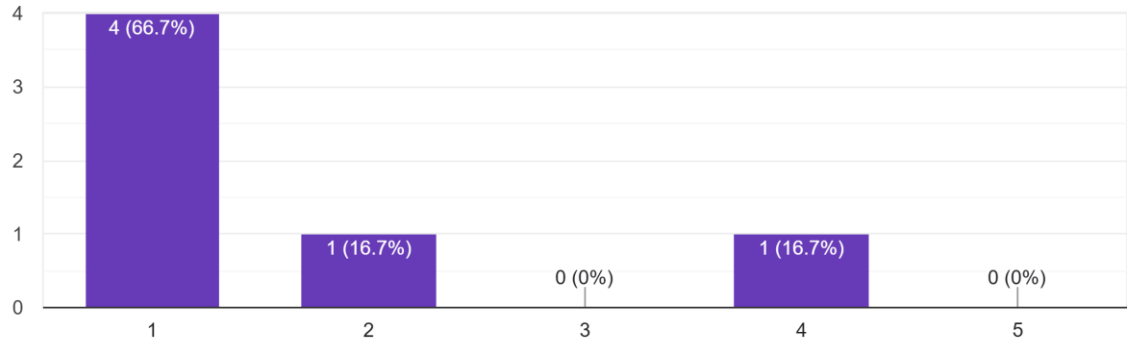
I thought the app was easy to use.

6 responses



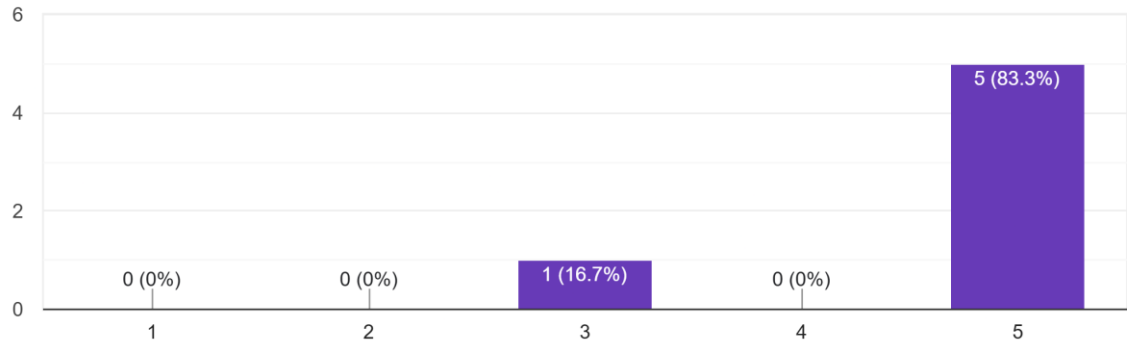
I think that I would need the support of a technical person to be able to use this app.

6 responses



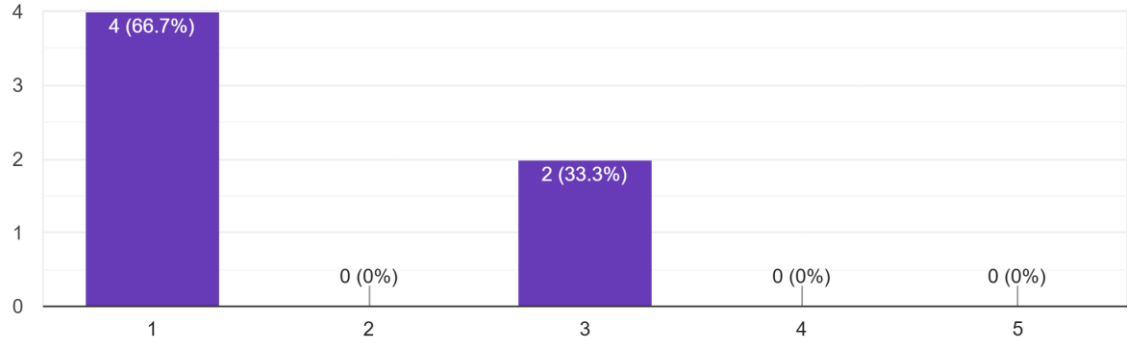
I found the various functions in this app were well integrated.

6 responses



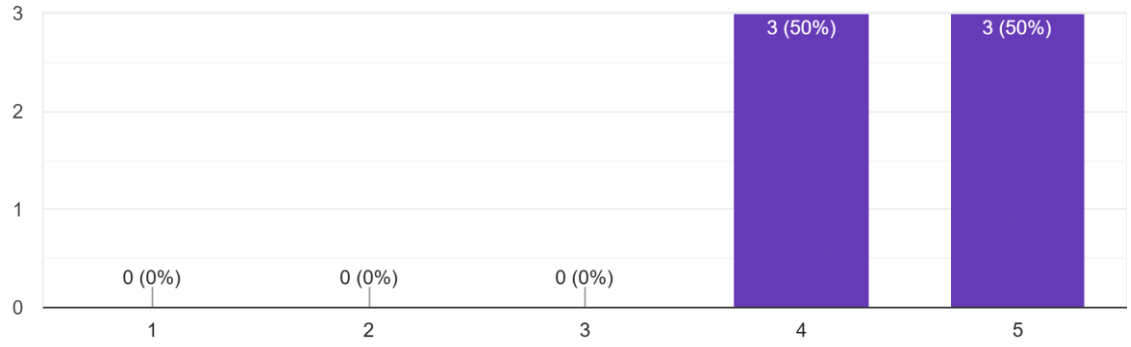
I thought there was too much inconsistency in this app.

6 responses



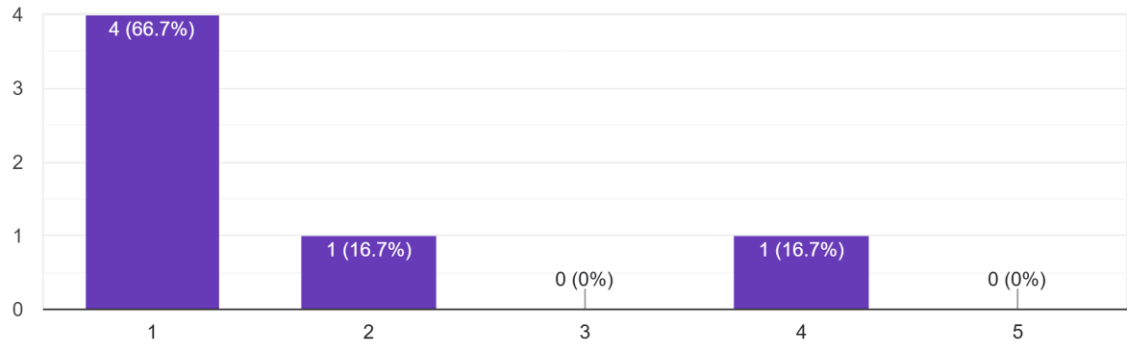
I would imagine that most people would learn to use this app very quickly.

6 responses



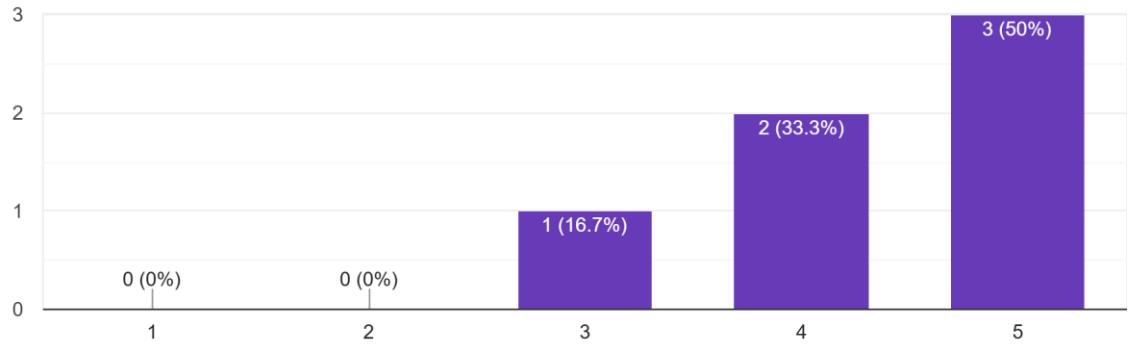
I found the app very cumbersome/awkward to use.

6 responses



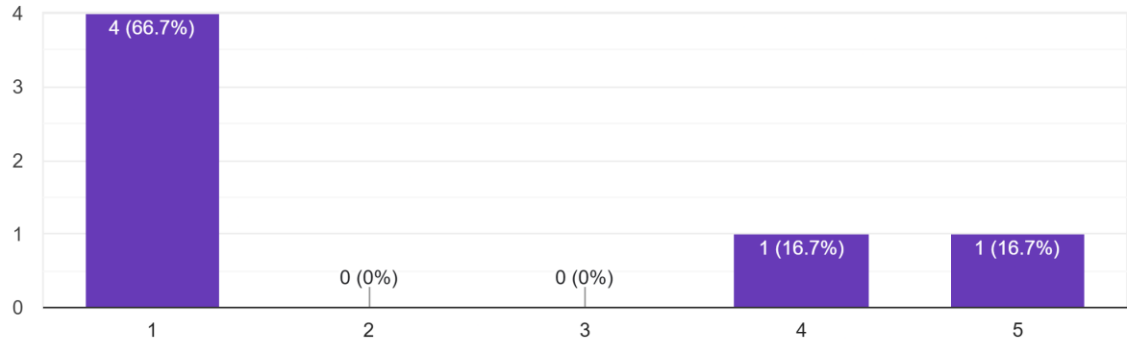
I felt very confident using the app.

6 responses



I needed to learn a lot of things before I could get going with this app.

6 responses



What was the most confusing part of the navigation?

4 responses

Nothing

At the moment, I feel okay. :)))

It was easy to use and quite intuitive.

i dont feel confused

How did you feel about the calibration process without the physical wristband?

5 responses

Interesting

good

I think it's fine if it's not have, but it's even better if it is.

Would be better if the band was available too but it was alright.

feel uncomfortable

What is one feature you would add or change to make the app better?

4 responses

Well, every feature is too good so i don't have any think to change this

Hmm at the moment i don't know yet :/

NA

Since u already have sign detection, I want to add the reverse capability: Text-to-Sign Translation. This feature takes paragraphs of text and visualizes them using a 3D hand model. This effectively turns our application into a complete two-way translation tool for the deaf community